UPM
UNIVERSITI PUTRA MALAYSIA
BERILMU BERBAKTI

INSTITUT PENYELIDIKAN PENUAAN MALAYSIA (MyAgeing®)

MALAYSIAN RESEARCH INSTITUTE ON AGEING

NCA2026

NATIONAL CONFERENCE ON AGEING 2026



Intersectionality and Inequities in Later Life:
Advancing Health, Social and
Technological Solutions for Ageing

ABSTRACT BOOK

19 – 21 OCTOBER 2026

KLGCC CONVENTION CENTRE
BUKIT KIARA, KUALA LUMPUR, MALAYSIA



TABLE OF CONTENTS

A : LIST OF ORAL PRESENTATION ABSTRACTS

1. KIDNEY - BRAIN AXIS IN OLDER ADULTS WITH TYPE 2 DIABETES: INFLAMMATION, ALBUMINURIA, AND CEREBROVASCULAR PERFUSION IN COGNITIVE-GAIT DECLINE
   Siddarth Raajasekar, Vasudevan Ramachandran, Pooja Ganesh, Mathiyazhagan Rajasekaran, Pandi Murugan Karuppaiah, & Jagadeeshwaran Venkataraman ……...1

2. CULTURAL AND AGE SENSITIVE GATEKEEPER TRAINING FOR OLDER Adults' MENTAL HEALTH: A PEER-TO-PEER MODULE DEVELOPMENT
   Asrenee Ab Razak, Fatin Afifah Rosli, Siti Nur Najibah Fauzi, Ahmad Shahril Ab Halim, A'isyah Che Rahimi, Nabilah Abdul Rahman, & Syaheedatul Iman Dinsuhaimi……...*2*

3. FROM FRACTURE TO FINANCIAL BURDEN: THE SOCIETAL COST OF HIP FRACTURE IN MALAYSIA
   Sivaraj Raman,Timothy Cheng Tsin Jien, Ahmad Luqman Abd Jamil, Khairatun Hisan Mohamad Amjadduzzahwi, & Gabriel Tang Ren Qian…….…….…3

4. EXPLORING PREPAREDNESS AND RESILIENCE AMONG FAMILY CAREGIVERS OF STROKE SURVIVORS
   Aniawanis Makhtar, Norliza Suut, Sharifah Munirah Syed Elias, & Norfadzilah Ahmad....….4

5. A SYSTEMATIC LITERATURE REVIEW ON FLEDGLING ADAPTATION AND FLEXIBILITY DESIGN STRATEGIES FOR AGE-FRIENDLY HOSPITALS
   Nurul Hanani binti Che Mohd Zain, Wan Norisma binti Wan Ismail, Madihah binti Mat Idris, & Fadli bin Arabi…....……...5

6. FACTORS ASSOCIATED WITH FUNCTIONAL LIMITATIONS IN ACTIVITIES OF DAILY LIVING AMONG OLDER ADULTS IN MALAYSIA
   Aini Fazana Zainal Abidin, & Raudhatul Mahfuzah Shafiai……..……6

7. LIVING ARRANGEMENTS AND LONG-TERM CARE FOR OLDER ADULTS IN THE ASEAN REGION: A SCOPING REVIEW PROTOCOL
   Nur Hidaah Mahamad Rappek, Asmidawati Ashari, & Mohammad Mujaheed Hassan……7

8. DEVELOPING THE CARE FRAMEWORK: A HUMAN ECOLOGY APPROACH TO SUPPORTING MALAYSIA'S SANDWICH GENERATIONS IN AN AGEING SOCIETY
   Yeshulavani Rajeswaran, Mohammad Mujaheed Hassan, Azlina Mohd Khir, & Asmidawati Ashari…….…….…8

9. GERONTOLOGIAL PERSPECTIVES ON DIGITAL USE: PATTERNS AND EXPERIENCES OF SMARTPHONE USE AMONG OLDER WOMEN AGED 60-80 IN MALAYSIA
Datin Kaysha Ubrani……...9

10. BEYOND THE MEDICAL MODEL: PRELIMINARY FINDINGS ON SOCIOCULTURAL PERSPECTIVES OF DEMENTIA CARE IN MALAYSIA
Nur Atiqah Azman, Wei Qi Koh, & Tan Maw Pin ……10

11. AGITATION IN ALZHEIMER'S DISEASE: TRANSLATING NEUROBIOLOGICAL INSIGHTS INTO EVIDENCE-BASED CLINICAL PRACTICE
Nur Hafidah Ishak……..11

12. EFFECTIVENESS OF A SELF-CARE COPING STRATEGIES INTERVENTION TO REDUCE DISTRESS AMONG DEMENTIA CAREGIVERS: A FEASIBILITY STUDY
Jeevasulochana Sinniah, Siti Anom Ahmad, & Halimatus Sakdiah Minhat……12

13. LONGITUDINAL TRANSITIONS IN SOCIAL FRAILTY STATES AMONG COMMUNITY-DWELLING OLDER ADULTS IN MALAYSIA Nur Aisyah Abdul Rahim, Choo Wan Yuen, Noran Naqiah Mohd Hairi, Halimah Awang, & Norma Mansor…….……..13

14. ASSOCIATION BETWEEN SOCIAL VULNERABILITY AND FRAILTY AMONG MIDDLE-AGED AND OLDER ADULTS IN MALAYSIA
Nur Azam Md Yusof, Noran Naqiah Mohd Hairi, Choo Wan Yuen, & Tan Chin Lung……...14

15. BEYOND THE ELDERLY CARE CENTRE: DESIGNING ACTIVE AGEING ENVIRONMENTS FOR PARTICIPATION, CONNECTION AND CONTRIBUTION Nur Atiqah Azman, Wei Qi Koh, & Tan Maw Pin……...15

16. BRIDGING GENERATIONS: OLDER ADULTS' PERSPECTIVES ON AN INTERGENERATIONAL PROGRAM WITH YOUNG CHILDREN
Siti Noraini Asmuri, Iylia Dayana Shamsudin, & Halimatus Sakdiah Minhat…….16

17. FOSTERING OLDER ADULT WELL-BEING: QUALITATIVE INSIGHTS INTO INTERGENERATIONAL ACTIVITY DESIGN IN MALAYSIA
Iylia Dayana Shamsudin, Siti Noraini Asmuri, & Halimatus Sakdiah Minhat…….17

18. FUNCTIONAL IMPAIRMENT IN LATER LIFE: MULTIDIMENSIONAL ASSOCIATIONSWITH NUTRITIONAL, ANTHROPOMETRIC AND LIFESTYLE FACTORS
Joanne Hui Xin Ang, & Yoke Mun Chan…….18

19. MEDICALLY TAILORED MEAL DELIVERY INTERVENTION IN VULNERABLE FOODINSECURITY ELDERLY PATIENTS ON HAEMODIALYSIS IN KLANG VALLEY, MALAYSIA: A FEASIBILITY PILOT STUDY
Bee Swen Yeen, Chan Yoke Mun, Juju Nakasha binti Jaafar, & Yang Wai Yew..……...… 19

20. COMPARATIVE PERFORMANCE OF WAIST CIRCUMFERENCE AND BODY MASS INDEX FOR SARCOPENIC OBESITY SCREENING AMONG MALAYSIAN OLDER ADULTS WITH HYPERTENSION
Kai Xin Lu, & Yoke Mun Chan……...…..20

21. MODALITY-SPECIFIC EFFECTS OF EARLY DARK-PHASE EXERCISE ON MUSCULAR AND BEHAVIOURAL OUTCOMES IN RATS WITH $ALCL_3$-INDUCED NEURODEGENERATION
Arimi Fitri Mat Ludin, Muhammad Hafiz Zuhdi Fairof, Amirul Hafiz Ahmad Abdullah, Nur Aisyah Anuar, Theng Choon Ooi, Farah Wahida Ibrahim, & Nor Fadilah Rajab…….21

22. IN VIVO GLYCEMIC SUPPRESSION BY PORANG GLUCOMANNAN (AMORPHOHALLUS MUELLERI) STARCH MATRICES: ESTABLISHING klu FOUNDATION FOR LOW-GI DYSPHAGIA BIGELS)
Amira Hasnanuha Ulayya, & Norhasnida Zawawi…….…..22

23. RANDOMIZED CONTROLLED HUMAN TRIAL ON THE EFFECT OF EDIBLE BIRD NEST SUPPLEMENTATION FOR SKIN AND CARDIOVASCULAR PROFILE IN PREMENOPAUSAL WOMEN: A PILOT STUDY
Salwana Ahmad, Kim Wei Chan, Der-Jiun Ooi, Norsharina Ismail, Norazlinaliza Salim, & Md Zuki Abu Bakar...……..……..23

24. INSIGHTS INTO DIGITAL MENTAL HEALTH READINESS AMONG EMOTIONALLY DISTRESSED OLDER ADULTS IN KELANTAN: A PRELIMINARY FINDING
Nurul Afiqah Che Mohd Zaid, Ahmad Shahril Ab. Halim, Wan Nor Arifin Wan Mansor, & Asrenee Ab Razak…….24

25. PSYCHOSOCIAL RISK FACTORS FOR DEPRESSION IN THE POST-RETIREMENT PERIOD: A CROSS-SECTIONAL STUDY OF MALAYSIAN GOVERNMENT PENSIONERS
Ahmad Shahril Ab. Halim, Raishan Shafini Bakar, & Sharifah Shakirah As-Sahab Wan Muhamad Nasir…….25

26. SUSTAINING GROUP SINGING IN DEMENTIA CARE: FACILITATION, ENVIRONMENT AND PERSON-CENTRED ADAPTATION IN A LONG-TERM QUALITATIVE DUAL-CASE STUDY IN GERMANY
Tan Teik Poi…….……..26

27. MAPPING QUALITY OF LIFE AND ASSOCIATED FACTORS AMONG COMMUNITY-DWELLING OLDER PERSONS USING SPATIAL AND NON-SPATIAL DATA IN SEPANG, SELANGOR: A STUDY PROTOCOL
Syaza Atikah Saidi Mohammad, Nor Afiah Mohd. Zulkefli, & Siti Aisah Mokhtar……27

28. INTEGRATING AGEING-IN-PLACE PRINCIPLES INTO MALAYSIA'S HOUSING DEVELOPMENT
Lina Emira Ahmad Adni, & Nor Zalina Harun…….……28

29. BRIDGING CARE NEEDS AND HOME ADAPTATION: HOME MODIFICATION AMONG FAMILY CAREGIVERS OF OLDER ADULTS IN MALAYSIA
Rahimah Ibrahim, Norisma Aiza Ismail, Mohd Fadzillah Bagat, Mohd Rizal Hussain, Ajun Chin, Siti Anom Ahmad, Sarifah Azizah Haron, Asmidawati Ashari, Shahrul Bahyah Kamaruzzaman, Roshanim Koris, & V Santha Vaithilingam……...……29

30. BRIDGING THE GAP IN GERIATRIC CARE: A MULTICENTER EVALUATION OF PSYCHOSOCIAL INEQUITIES AND CAREGIVER BURDEN IN MALAYSIAN TERTIARY HOSPITALS
Sivasangari Subramaniam, Thiam Chiann Ni, Mogana Thulaisi, Norshahida Binti Abdul Hamid, Keng Jun Ni, Ang Zi Heng, Adilah Aminuddin, Sai Deepikaa Sotheenathan, & Sotheenathan Krishinan……..…...……30

31. DEVELOPING THE SUCCESSFUL AGEING READINESS FRAMEWORK (SARF): A MIDLIFE APPROACH TO HEALTHY AND SUCCESSFUL AGEING IN MALAYSIA
Yeshulavani Rajeswaran, Mohammad Mujaheed Hassan, Azlina Mohd Khir, & Asmidawati Ashari…….……...……31

32. SUB-DISTRICT MAPPING OF AGED POPULATIONS IN MALAYSIA: LESSONS AND POTENTIAL
Chai Sen Tyng, Danesh Prakash Chacko, Rahimah Ibrahim, Muhammad Mujaheed Hassan, & Abdul Rashid Mohamed Shariff…….……...……32

B : LIST OF POSTER PRESENTATION ABSTRACTS

1. POPULATION AGEING AND LONGEVITY IN MALAYSIA: TRENDS IN MORTALITY AND EXPECTED RETIREMENT DURATION
Halim Shukri Kamaruddin……..…….……...……...……34

2. NAVIGATING THE DEFICIT AND NOSTALGIA VIEW TO STRATEGIC VALUE OF AGEING WORKFORCE USING A FOUR-QUADRANT MODEL FOR SUSTAINABLE AI-HRM
Low Mei Peng, Har Wai Mun, & Ting Li Han……..35

3. LOCALISING DIGITAL LITERACY TRAINING FOR OLDER PERSONS IN INDONESIA: A TRAINING NEEDS ANALYSIS THROUGH STAKEHOLDER FOCUS GROUP DISCUSSION
Siti Farra Zillah Abdullah, Mohamad Fazdillah Bagat, Siti Anom Ahmad, Farahdibha Tenrilemba, Fakhrul Zaman Rokhani, & Nur Ainani Mahyuddin……36

4. FORMAL EDUCATION AND LIFELONG LEARNING FOR HEALTHY AGEING IN SOUTH EAST ASIA
Maria Alaguru……37

5. FAMILY CAREGIVING FOR OLDER ADULTS: BALANCING CAREGIVING CHALLENGES AND POSITIVE EXPERIENCES
Norisma Aiza Ismail & Rahimah Ibrahim…….38

6. IN VITRO HEPATO-PROTECTIVE OF MIXED BERRIES POWDER, BET-AGING™
Che Muhammad Khairul Hisyam Bin Ismail, Mohammad Adi Bin Mohammad Fadzil, Suraiami Binti Mustar, Mohd Naeem Bin Mohd Nawi, Noor Farhana Binti Ab Karim, & Aswir Bin Abd Rashed……..……39

7. WHAT MAKES GOOD DEMENTIA CARE? PERSPECTIVES OF PERSONS WITH DEMENTIA AND FAMILY CAREGIVERS
Sharifah Munirah Syed Elias, Nur Fatihah Shuhada Ismail, Aniawanis Makhtar, Mohd Aznan Md Aris, & Mohd Zulkifli Mohamad Zahir…….....40

8. MUSCLE POWER MATTERS: ASSOCIATIONS WITH FRAILTY, PHYSICAL PERFORMANCE, AND BODY COMPOSITION AMONG OLDER PERSONS
Raja Nurzatul Efah Raja Adnan, Hazwan Mat Din, Siti Nor Hadirah Azman, Najwa Najihah Mohd Azmi, Ajun Chin, & Nur Hafidah Ishak.……41

9. DEVELOPMENT, VALIDATION AND EVALUATION OF A MULTIDISCIPLINARY INTEGRATED CARE PROTOCOL FOR THE IDENTIFICATION AND MANAGEMENT OF MALNUTRITION-SARCOPENIA SYNDROME IN HOSPITALIZED OLDER ADULTS: A STUDY PROTOCOL
Aliza Haslinda Hamirudin, Saminur Fauzan, Siti Salwana Kamsan, Wan Fathin Fariza Wan Mahmood, Nor Syuhaidah Udin, & Mohd Zulkifli Mohamad Zahir..…….42

10. PSYCHOLOGICAL AND DIGITAL DETERMINANTS OF MHEALTH ADOPTION AMONG OLDER ADULTS: THE MEDIATING ROLE OF SELF-EFFICACY
Golaleh karbasi, Rahimah Ibrahim, Siti Anom binti Ahmad, Ghobad Moradi, Puvaneswaran Kunasekaran, Nur Hafidah Ishak, & Mohd Nazim Mohtar…..……...…….43

11. TOWARDS A NATIONAL AGE-FRIENDLY HOSPITAL FRAMEWORK FOR MALAYSIA
Zahira Zohari, Hakimah Sallehudin, Rahimah Ibrahim, & Tan Maw Pin……..……..44

12. SELF-RATED HEALTH AS A MEDIATOR OF INPATIENT VISITS AMONG OLDER ADULTS
Nurul Salwana Abu Bakar, Aidalina Mahmud, & Lim Poh Ying……...……..……45

13. MILD BEHAVIOURAL IMPAIRMENT AS A PREDICTOR OF DEMENTIA: A SYSTEMATIC REVIEW OF CLINICAL, NEUROBIOLOGICAL, AND PSYCHOSOCIAL EVIDENCE (2016–2026)
Ahmad Shahril Ab. Halim, Norili Farhana Ahmad Saberi, Asma Parveen, Heba Almassri, & Asrenee Ab Razak……..…….....……...46

14. BRIDGING CLINIC AND COMMUNITY: A CASE STUDY ON HEARING SCREENING INITIATIVES FOR OLDER ADULTS
Mohd Fadzil Nor Rashid, Aw Cheu Lih, Mahamad Almyzan Awang, Maziah Romli, Mohd Normani Zakaria, Wan Najibah Wan Mohamad, & Hasrul Hosshan……..…….47

15. NEUROPROTECTIVE EFFECTS OF MORINGA OLEIFERA CHEWABLE TABLETS ON COGNITIVE DECLINE IN STZ-INDUCED SPRAGUE-DAWLEY RATS: EVIDENCE FROM Y-MAZE BEHAVIORAL ASSESSMENT
Yus Aniza Yusof, Nurjuliana Mokhtar, Nur ‘Aliaa Abd. Rahman, Hafandi Ahmad, Aswir Abd Rashed, Mohammad Adi Mohammad Fadzil, Muhamad Faiz Othman, Wan Hamirul Bahrin Wan Kamal, Saifullizan Mohamad, Anee Suryani Sued, Siti Suhaila Md Izah, Ng Yen, Nur Fariza Mohd Kamil, Siti Aminah Muhamad, Hishar Hassan, & Syaiful Hizam Kamaruddin.48

16. SILENT MARITAL DISTRESS, VISIBLE POSSESSION: SAKA AS A CULTURAL IDIOMOF DISTRESS IN AN ELDERLY MALAY WOMAN
Nabilah Abdul Rahman, & Asrenee Ab Razak……..49

17. EFFECTIVENESS OF THE LOW INTENSITY, REPETITIVE, MULTIMODAL WALKING EXERCISE (GOLDEN STEPS) ON PHYSICAL ACTIVITY LEVEL AMONG THE ELDERLY
Nur Syahirah Mohamad Rosli, Noorismiza Zainudin, Siti Nur Baait Sokran, Muslimah Ithnin, Sakinah Rasli, Nur Hafini Harith, & Nurul’ain Shuhaimi……50

18. VIRTUAL REALITY-BASED RELIGIOUS TOURISM AND PSYCHOLOGICAL WELLBEING AMONG MUSLIM OLDER PERSONS: A MIXED-METHODS STUDY PROTOCOL
Mohd Rizal Hussain, Mohd Nazim Mohtar, Nur Hafidah Ishak, Raja Nurzatul Efah Raja Adnan, Ajun Chin, Norhalina Senan, Noorhaniza Wahid, Rahayu A. Hamid, Norhanifah Murli, Che Samihah Che Dalim, Suriawati Suparjoh, & Muhammad Fakri Othman..……51

19. SIMAGEING: PREDICTIVE SPATIAL CAD MODELLING AND AGENT-BASED BEHAVIOURAL ANALYTICS FOR NEXT-GEN MHIT UNDERWRITING
Ferozan Ahmad, & Mohd Nazim Mohtar..……52

ABSTRACT BOOK

01

Oral Presentation Abstracts

19–21 October 2026

KLGCC Convention Centre, Kuala Lumpur

KIDNEY - BRAIN AXIS IN OLDER ADULTS WITH TYPE 2 DIABETES: INFLAMMATION, ALBUMINURIA, AND CEREBROVASCULAR PERFUSION IN COGNITIVE-GAIT DECLINE

Siddarth Raajasekar, Vasudevan Ramachandran, Pooja Ganesh, Mathiyazhagan Rajasekaran, Pandi Murugan Karuppaiah, & Jagadeeshwaran Venkataraman

Abstract

Background: Chronic inflammation, metabolic dysregulation, and microvascular injury associated with type 2 diabetes mellitus (T2DM) contribute to cognitive decline and functional impairment in older adults. **Methods:**This study aimed to investigate the kidney–brain axis by examining the associations between inflammatory biomarkers, albuminuria, cerebrovascular perfusion, cognitive performance, and gait function in older adults with T2DM. A retrospective cross-sectional secondary analysis was conducted using data from the publicly available GE-75 cohort, originally collected at Beth Israel Deaconess Medical Centre, Boston, Massachusetts, USA. The present study included 88 community-dwelling older adults aged 50–85 years with T2DM (median age: 66 years, interquartile range: 60 - 72 years). Correlation and regression analyses were performed to evaluate the relationships the variables. **Results:** Albumin-creatinine ratio demonstrated a negative association with global cerebrovascular reactivity ($r = -0.261$, $p = 0.087$; $n = 44$) and six-minute walk distance ($r = -0.251$, $p = 0.082$; $n = 49$), suggesting that greater renal microvascular injury was associated with reduced cerebrovascular reserve and functional capacity. Interleukin-6 showed a moderate positive correlation with frontal lobe perfusion ($r = 0.334$, $p = 0.175$; $n = 18$) and a negative correlation with Mini-Mental State Examination score ($r = -0.269$, $p = 0.227$; $n = 22$). Glycated haemoglobin (HbA1c) demonstrated a weak positive association with global cerebrovascular reactivity ($r = 0.214$, $p = 0.158$; $n = 45$). **Conclusion:** Albuminuria emerged as a promising biomarker of the kidney–brain axis, linking renal microvascular injury with cerebrovascular dysfunction and functional decline in older adults with T2DM. These findings support integrated, multi-domain biomarker assessment to identify individuals at increased risk of cerebrovascular and cognitive complications and warrant validation in larger prospective studies.

Keywords: Older Adults; Type 2 Diabetes Mellitus; Kidney–Brain Axis; Albuminuria; Cerebrovascular Perfusion; Cognitive Function.

CULTURAL AND AGE SENSITIVE GATEKEEPER TRAINING FOR OLDER ADULTS' MENTAL HEALTH: A PEER-TO-PEER MODULE DEVELOPMENT

Asrenee Ab Razak, Fatin Afifah Rosli, Siti Nur Najibah Fauzi, Ahmad Shahril Ab Halim, A'isyah Che Rahimi, Nabilah Abdul Rahman, & Syaheedatul Iman Dinsuhaimi

Abstract

Introduction: Population ageing has increased the burden of late-life mental health disorders, while help-seeking among older adults remains low due to stigma, generational beliefs, and communication barriers. Community-based strategies that leverage trust within the same age group may improve early detection and support. A culturally and age-sensitive peer gatekeeper model, where retirees assist fellow older adult individuals, may bridge gaps between community members and formal services. **Objective:** To develop and validate a culturally and age-sensitive elderly mental health gatekeeper training module using a peer-to-peer approach among retirees in Malaysia. **Method:** A multi-phase module development study was conducted. Initial content was derived from literature review, ageing psychology frameworks, and community consultations. The module covered recognition of emotional distress, role of gatekeepers, psychological first aid skills and stigma. Cultural adaptation incorporated spirituality, collectivist values and storytelling pedagogy. Content validation employed a modified Delphi method involving experts in psychiatry, clinical psychologist, family physician and community health across two iterative rounds. Items were rated for relevance and clarity, and the Content Validity Index (CVI) was calculated at item (I-CVI) and scale (S-CVI) levels. Retiree volunteers then underwent pilot training with role-play and simulation exercises. **Results:** Consensus was achieved after two Delphi rounds. The module demonstrated excellent content validity (I-CVI range: 0.80-1.00; S-CVI/Ave: 0.92). Experts recommended emphasising peer trust, culturally respectful communication, and simplified referral guidance. Pilot participants reported improved confidence in recognising emotional distress and providing supportive conversations. The peer-to-peer approach enhanced openness among older adult participants and reduced stigma toward mental health discussion. **Conclusion:** The culturally adapted peer gatekeeper module is valid and acceptable for community implementation. Training retirees as lay supporters strengthens early identification of distress and facilitates linkage to care. This approach represents a scalable community mental health strategy for ageing populations in culturally diverse settings**.**

Keywords: Cultural, Training, Older, Mental health, Peer to Peer,

FROM FRACTURE TO FINANCIAL BURDEN: THE SOCIETAL COST OF HIP FRACTURE IN MALAYSIA

Sivaraj Raman,Timothy Cheng Tsin Jien, Ahmad Luqman Abd Jamil, Khairatun Hisan Mohamad Amjadduzzahwi, & Gabriel Tang Ren Qian

Abstract

Fragility hip fractures are a major consequence of population ageing, leading to substantial healthcare utilisation, premature mortality, and productivity losses. This study estimated the one-year societal economic burden of fragility hip fractures in Malaysia in 2025 and identified key cost drivers to inform investment in preventive strategies. An incident-based cost-of-illness model from a societal perspective was developed, using age- and sex-specific incidence and mortality from published Malaysian data and national population statistics. In the absence of comprehensive local cost data, a cross-sectional study at a public tertiary hospital provided mean cost parameters applied to national case estimates. Retrospective medical record abstraction determined healthcare provider costs for surgery, implants, inpatient management, outpatient care, and rehabilitation, while a prospective survey captured patient and carer out-of-pocket expenditures and productivity losses. Productivity losses due to absenteeism and premature mortality were valued using the human capital approach. Other relevant cost components were obtained from local literature. Deterministic sensitivity analyses varied key epidemiological and cost parameters to explore robustness of the model. It was estimated that a total of 7,576 new fragility hip fractures occurred, resulting in 1,109 deaths within one year. The total one-year societal economic burden was RM256.6 million, corresponding to a mean cost of approximately RM33,800 per case. Inpatient care cost was the largest component at RM90.7 million (35.4%), followed by patient and carer productivity losses (RM63.0 million, 24.6%), informal care cost (RM62.2 million, 24.2%), out-of-pocket expenditures (RM27.8 million, 10.8%), and outpatient healthcare (RM12.9 million, 5.0%). Sensitivity analysis showed hip-fracture incidence as the most influential driver of total cost, followed by the proportion undergoing surgery. Fragility hip fracture imposes a substantial societal burden beyond hospital costs, with patients, carers, and productivity losses driving more than half of total expenditures, underscoring the need to prioritise prevention and integrated post-fracture care in Malaysia's ageing population.

Keywords: Hip Fracture, Economic Burden, Productivity Loss, Cost Saving

EXPLORING PREPAREDNESS AND RESILIENCE AMONG FAMILY CAREGIVERS OF STROKE SURVIVORS

Aniawanis Makhtar, Norliza Suut, Sharifah Munirah Syed Elias, & Norfadzilah Ahmad

Abstract

Background: Family caregivers play a central role in supporting stroke survivors after hospital discharge, yet many assume complex caregiving responsibilities with limited preparation. **Objective:** This study explored how family caregivers experienced preparedness and resilience while caring for stroke survivors in Malaysia. **Methods:** This qualitative study used an interpretive descriptive approach. Twenty-six family caregivers of stroke survivors were recruited from a tertiary hospital, community-based stroke support and rehabilitation centres, and online platforms in Malaysia. Semi-structured interviews were conducted to explore caregivers' experiences of preparedness, resilience, and support. Data were analysed using reflexive thematic analysis. **Results:** Five themes were constructed from the data. First, caregiving began before readiness could be established, reflecting the abrupt transition into unfamiliar responsibilities. Second, readiness developed through uneven access to knowledge and resources, with practical caregiving confidence shaped by the availability of usable information and material support. Third, being strong meant continuing despite emotional, physical, and social strain, showing that resilience did not mean being unaffected by caregiving demands. Fourth, acceptance transformed caregiving disruption into a meaningful responsibility through emotional adjustment, filial obligation, and spiritual meaning-making. Fifth, caregiving responsibility was experienced individually but sustained collectively through trusted relationships, experiential knowledge from peers, and community support. Across the themes, caregivers described exhaustion, financial strain, social restriction, inconsistent preparation, and gradual adaptation to the caregiving role. **Conclusion:** Preparedness and resilience were not fixed qualities but evolving processes shaped by caregiving demands, access to knowledge and resources, personal meaning, and social support. Findings highlight the need for early, practical, and context-sensitive caregiver preparation that extends beyond hospital discharge. Healthcare services should strengthen caregiver education, follow-up support, emotional care, and access to community and material resources to better sustain family caregivers throughout the stroke caregiving journey. Such support may strengthen caregivers' confidence while reducing preventable strain during transitions from hospital to home**.**

Keywords: Preparedness, Resilience, Caregivers, Stroke, Survivor

A SYSTEMATIC LITERATURE REVIEW ON FLEDGLING ADAPTATION AND FLEXIBILITY DESIGN STRATEGIES FOR AGE-FRIENDLY HOSPITALS

Nurul Hanani binti Che Mohd Zain, Wan Norisma binti Wan Ismail, Madihah binti Mat Idris, & Fadli bin Arabi

Abstract

Ageing population is increasing demand for hospital environments that are safer, supportive, and functionally responsive to the needs of older adults, thereby necessitating evidence-based adaptation and flexibility design strategies within the facilities. However, research addressing how hospitals can incorporate age-friendly design principles remains fragmented across disciplines. Thus, this Systematic Literature Review (SLR) aims to identify, synthesise, and categorise the evidences on adaptation and flexibility design strategies for Age-Friendly Hospitals (AFHs). The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol using two major databases, Scopus and Web of Science (WoS). Advanced searching strategies were applied using the keywords "adaptation," "flexibility," "design strategy," "age-friendly," "hospital," and "healthcare facility," leading to the selection of 57 primary studies (n = 57) from the articles of the year 2025 for final synthesis. The findings revealed that the adaptation and flexibility strategies in AFH design are commonly discussed through three dominant thematic domains. First, Falls, Frailty, Functional Risk, and Patient Safety in Older Adults which highlighted the prevention of risks and functional provision through architectural modification in hospital environments. Second, Digital Health, Health Technology, Measurement Tools, and Clinical Implementation emphasised the increasing role of health technologies, validated assessment tools and clinical implementation models to aid the care and monitoring of older adults. Third, Care Experience, Adaptation, and Continuity of Care (Hospital-Home-Community) that concerns of transitional care, support of caregivers, integration of services, and alignment of the hospital environments with the community-based ageing requirements. The synthesis suggests that AFH change implementation needs a multidimensional strategy involving physical design flexibility, built environment interventions aimed at safety, and continuity of care across settings aided by technology. They are vital to provide an informed decision for future research on hospital designs that support age-responsive hospital settings.

Keywords: Adaptation, Flexibility, Design Strategy, Age-friendly, Hospital, Healthcare Facility

FACTORS ASSOCIATED WITH FUNCTIONAL LIMITATIONS IN ACTIVITIES OF DAILY LIVING AMONG OLDER ADULTS IN MALAYSIA

Aini Fazana Zainal Abidin, & Raudhatul Mahfuzah Shafiai

Abstract

Despite the growing ageing population, research examining the role of social support in functional limitations among older adults remains scarce. Understanding how social support influences the ability of older persons to perform daily activities is important for informing healthcare planning, resource allocation, and the development of policies that promote healthy ageing and reduce caregiving burden. Accordingly, this study aimed to investigate the influence of social support and selected sociodemographic characteristics on limitations in activities of daily living among older persons in Malaysia. Preliminary data from the Sixth Malaysia Population and Family Survey (MPFS-6), a nation-wide, cross-sectional survey using a two-stage stratified sampling design, were analyzed. A representative sample of individuals aged ≥60 years identified through national sampling frames throughout Malaysia was included. Face-to-face interviews were conducted using validated questionnaires on basic activities of daily living (ADL) and instrumental activities of daily living (IADL). Social support was measured using the Duke Social Support Index. The overall prevalence of the presence of any limitations in ADL and IADL was 11.6% and 34.2% respectively. Overall prevalence of the older persons with low social support was 56.4% and high social support was 43.6%. Multinomial Logistic Regression analysis identified the following factors as being associated with limitations in ADL and IADL: being female, of older age, having a monthly income, Female sex, older age, low social support, and low income were consistently associated with both ADL and IADL limitations among older adults. Interventions aimed at enhancing social support and reducing socioeconomic disadvantage may help preserve functional independence and improve quality of life in later life.

Keywords: Older Adults, Social Support, Activities of Daily Living (ADL), Instrumental Activities of Daily Living (IADL), Malaysia

LIVING ARRANGEMENTS AND LONG-TERM CARE FOR OLDER ADULTS IN THE ASEAN REGION: A SCOPING REVIEW PROTOCOL

Nur Hidaah Mahamad Rappek, Asmidawati Ashari, & Mohammad Mujaheed Hassan

Abstract

Background: The ASEAN region is undergoing rapid demographic aging, placing unprecedented pressure on traditional informal care networks and formal long-term care (LTC) infrastructure. Understanding how living arrangements interact with LTC delivery systems is critical for developing sustainable eldercare policies across diverse socio-economic contexts. **Objective**: This scoping review aims to systematically map existing empirical literature and national policy frameworks governing living arrangements and long-term care systems for older adults across ASEAN. **Methods**: Three databases namely Scopus, PubMed, and SpringerLink with guided by the Joanna Briggs Institute (JBI) methodology and the Population, Concept and Context (PCC) framework were applied to establish the inclusion criteria of the scoping review as follows: Population: Older adults (aged 60 and above) and pre-older cohorts (aged 50years and above), Concept: Living arrangements (co-residence, living alone, institutional setups) and Long-Term Care (formal, informal) and Context: The ASEAN region (Malaysia, Thailand, Indonesia, Singapore. Philippines, Vietnam, Myanmar, Brunei, Cambodia, Laos, Timor-Leste). The PRISMA-ScR Checklist guidelines was applied for transparent reporting. Peer-reviewed empirical articles, policy documents, and reviews published in English were included. Studies were screened independently by title, abstract and full text, with data mapped narratively and structurally charted. **Results & Significance**: Of the 5,765 articles identified, 5,550 titles, 583 abstracts, and 341 full-text articles were screened for eligibility. The review included 116 studies. Descriptive patterns indicate a heavy concentration of empirical literature in Singapore, Indonesia, Malaysia and Thailand. This review identifies research and policy gaps across ASEAN countries and clarifies structural drivers of living arrangements and LTC utilization. While traditional home and family-based care remain preferred due to filial piety norms, interest in formal living arrangements such as Retirement Villages and non-profit LTC homes is growing. **Conclusion**: The findings provide a critical baseline to inform eldercare policy frameworks and justify empirical investigations into future care intentions among pre-retirement populations.

Keywords: Living Arrangements, Long-Term Care, Older Adults, ASEAN, Scoping Review

DEVELOPING THE CARE FRAMEWORK: A HUMAN ECOLOGY APPROACH TO SUPPORTING MALAYSIA'S SANDWICH GENERATIONS IN AN AGEING SOCIETY

Yeshulavani Rajeswaran, Mohammad Mujaheed Hassan, Azlina Mohd Khir, & Asmidawati Ashari

Abstract

Introduction: Malaysia's transition towards an ageing society has intensified the demand for family-based caregiving, placing increasing responsibilities on middle-aged adults who simultaneously care for ageing parents and dependent children. Commonly referred to as the sandwich generation, these individuals play a critical role in sustaining informal care systems while balancing employment, financial obligations, and family commitments. However, caregiving remains largely under-recognised within policy and service planning despite its implications for caregiver wellbeing, workforce productivity, family resilience, and long-term care sustainability. This paper aims to develop the CARE Framework as a conceptual model to guide a more coordinated and comprehensive approach to supporting Malaysia's sandwich generation. **Methods:** A narrative literature review and thematic synthesis were undertaken using peer-reviewed journal articles, policy documents, and international caregiving frameworks published between 2000 and 2025. Literature relating to caregiving, caregiver burden, sandwich generation experiences, family support, workplace interventions, and ageing policy was critically examined. Human Ecology Theory was adopted as the guiding theoretical perspective to conceptualise caregiving as a multidimensional phenomenon shaped by interactions across individual, family, workplace, community, and policy systems. **Results:** The review identified five interrelated domains that influence caregiver wellbeing and caregiving capacity: caregiver health and wellbeing, family and social support, workplace support systems, financial and service assistance, and policy and community infrastructure. These domains informed the development of the CARE Framework, which proposes an integrated ecosystem of support designed to strengthen caregiver resilience, enhance caregiving capacity, and promote sustainable family-based care. **Discussion:** The CARE Framework contributes to ageing and caregiving scholarship by repositioning caregivers as a strategic population within Malaysia's ageing agenda rather than as an invisible family resource. The framework provides a foundation for future empirical validation, policy development, and multi-sector interventions aimed at strengthening family resilience, supporting productive ageing, and enhancing the sustainability of long-term care in an ageing society.

Keywords: Ageing Workforces, AI-Enabled Sustainable HRM, Decent Work, SDG 8, SDG 10, Silver Economy

GERONTOLOGIAL PERSPECTIVES ON DIGITAL USE: PATTERNS AND EXPERIENCES OF SMARTPHONE USE AMONG OLDER WOMEN AGED 60-80 IN MALAYSIA

Datin Kaysha Ubrani

Abstract

Digital Technologies are increasingly embedded in later life, yet older adults' digital experiences are often interpreted through Western-centric and age0 deterministic assumptions. This study examined smartphone use aiming urban Malaysian women aged 60-80, with particular attention to digital engagement, independence and psychosocial well-being. A sequential explanatory mixed-methods design was used, comprising an online survey (n=29) followed by semi-structured interviews (n=5). Quantitative analyses examined whether age cohort and daily smartphone usage were associated with behavioural patterns, digital confidence, perceived independence and emotional well-being. Qualitative interviews provide interpretive insight into participants' experiences of smartphone use. Quantitative findings indicated no significant associations between chronological age and digital confidence, perceived independence or emotional well-being. Younger-old participants (60-70 years) reported higher levels of social-entertainment use, but otherwise age-related differences were limited. In contrast, daily smartphone usage appeared more strongly related to well-being: participants reporting four or more hours or daily use showed significantly higher perceived independence and emotional well-being. Qualitative findings provided further insight, conceptualising smartphones as a "double-edged lifeline" that supported autonomy, social connection and emotional well-being while also introducing new responsibilities and demands for self-regulation. Participants described purposeful, selective engagement with digital technologies, often using boundary setting to maintain control over their digital lives. Intergenerational support emerged as an important facilitator of positive digital experiences, particularly for learning and sustained engagement. Participants occupied a spectrum of user positions, from purposeful integrators to pragmatic controllers, challenging deficit-based assumptions that older adults are technologically homogenous or resistant. These findings suggest that meaningful engagement, rather than chronological age alone, may be more important in understanding digital experiences and psychosocial well-being in later life. The study highlights the importance of relational support, autonomy and purposeful digital engagement in promoting healthy ageing and digital inclusion among older adults in Malaysia.

Keywords: Older Women, Smartphone Use, Digital Engagement, Perceived Independence, Emotional Wellbeing, Mixed-Methods Research.

BEYOND THE MEDICAL MODEL: PRELIMINARY FINDINGS ON SOCIOCULTURAL PERSPECTIVES OF DEMENTIA CARE IN MALAYSIA

Nur Atiqah Azman, Wei Qi Koh, & Tan Maw Pin

Abstract

Background: Dementia care is shaped not only by the biomedical parameters but also by the biopsychosocial context, where social and cultural factors profoundly influence care delivery and reception. Malaysia's rich sociocultural diversity, characterized by distinct family dynamic structures, filial caregiving expectations, community stigma, and traditional health beliefs, shape daily dementia practices. However, these sociocultural nuances remain insufficiently explored from the lived perspectives of persons living with dementia (PLWD) and their caregivers. Understanding these viewpoints is to develop culturally responsive dementia care practices. **Objective:** To explore the perceptions of PLWD and family caregivers regarding how social and cultural factors influence dementia care practices in Malaysia. **Methods:** A qualitative study using purposive and snowball sampling was conducted among ten participants (n=10), comprising PLWD with mild-to-moderate dementia (receiving care for >1 year) and their primary family caregivers (>6 months caregiving duration). Semi-structured interview, guided by the Consolidated Framework for Implementation Research (CFIR), were conducted face-to-face, online, or via telephone. Audio-recorded Interviews were transcribed verbatim and analysed using thematic analysis in ATLAS.ti to identify patterns of sociocultural influences. **Results**: Preliminary analysis indicates that sociocultural contexts significantly shape dementia care experiences. Key emerging themes include the influence of filial caregiving obligations, gendered family roles, communication barriers, delayed help-seeking, spiritual and traditional health beliefs, and culturally embedded daily routines. Detailed qualitative themes and illustrative quotes will be presented during the conference. **Conclusion:** Lived experiences of PLWD and family caregivers provide essential insights into how sociocultural dynamics influence dementia care in Malaysia. These preliminary findings highlight the need to establish culturally grounded dementia care competencies for health workers.

Keywords: Family Caregivers, Dementia Care, Sociocultural, People Living with Dementia, Perception

AGITATION IN ALZHEIMER'S DISEASE: TRANSLATING NEUROBIOLOGICAL INSIGHTS INTO EVIDENCE-BASED CLINICAL PRACTICE

Nur Hafidah Ishak

Abstract

Neuropsychiatric symptoms (NPS) are core, defining features across all stages of Alzheimer's disease (AD). In a cross-sectional study of 435 patients with AD, 90% exhibited at least one NPS. Agitation and aggression represent particularly prevalent manifestations, affecting 63% of individuals across their disease course and 52% in the preceding month alone. Clinical Definition & Drivers: According to the International Psychogeriatric Association (IPA), agitation in cognitive disorders requires an underlying dementia syndrome, at least one observable agitated behavior, persistence or frequent recurrence for over two weeks, and severity sufficient to cause excess distress. Manifestations span three distinct domains: excessive motor activity (pacing, restlessness, repetitiveness, hoarding), verbal aggression (screaming, profanity, repetitive questioning), and physical aggression (hitting, kicking, biting, throwing objects). Agitation stems from fronto-limbic circuit dysfunction, neuroinflammation, and monoaminergic imbalance (dysregulation of noradrenergic, serotonergic, and dopaminergic networks). These neurobiological vulnerabilities are triggered by acute somatic factors (pain, infection, constipation, delirium, sleep disturbance, adverse drug effects), unmet needs (hunger, discomfort, fear, frustration, impaired communication), and environmental stressors (noise, sensory overload, disrupted routines, challenging caregiving dynamics). Clinical, Economic, & Caregiver Burden: Agitation in Alzheimer's disease (AAD) compounds the overall disease burden, accelerating disease progression, cognitive and functional decline, comorbidity accumulation, and mortality. In long-term care (LTC) settings, AAD precipitates a secondary cascade of falls, fractures, infections, polypharmacy, and emerging NPS. Economically, AAD markedly increases direct healthcare costs, hospitalizations, specialty consultations, and institutionalization rates. For informal caregivers, AAD is a primary driver of physical health decline, reduced quality of life, depression, anxiety, social isolation, feelings of guilt or embarrassment, and elevated healthcare utilization. Therapeutic Evolution: While non-pharmacological strategies serve as foundational first-line care, they often prove insufficient for severe agitation. Historically, clinicians relied on off-label typical and atypical antipsychotics, which offered modest benefit alongside severe adverse events, including excessive sedation, extrapyramidal symptoms (EPS), accelerated cognitive blunting, and increased mortality. In May 2023, the U.S. FDA approved brexpiprazole as the first targeted pharmacotherapy indicated specifically for agitation associated with dementia due to Alzheimer's disease. As a serotonin-dopamine activity modulator (SDAM), brexpiprazole combines partial agonism at dopamine D2 and serotonin 5-HT1A receptors with antagonism at serotonin 5-HT2A and noradrenergic alpha1B/alpha2C receptors, directly stabilizing aberrant monoaminergic transmission without inducing broad neuroinhibition. Efficacy, Safety, & Management Strategies: In 12-week Phase 3 randomized, placebo-controlled trials, brexpiprazole (target doses of 2 mg/day and 3 mg/day) achieved statistically significant, clinically meaningful reductions in agitation frequency on the Cohen-Mansfield Agitation Inventory (CMAI) and global disease severity on the Clinical Global Impression-Severity (CGI-S) scale versus placebo. Treatment-emergent adverse events were predominantly mild to moderate (headache, dizziness, urinary tract infection, somnolence), with sedation and EPS rates comparable to placebo. Optimal clinical management mandates systematic exclusion of reversible somatic and environmental triggers before prescribing, disciplined titration (initiating at 0.5 mg/day to a 2–3 mg/day target), and continuous safety monitoring regarding the class-wide boxed warning for mortality in dementia-related psychosis. **Conclusion:** Agitation in AD is a debilitating syndrome compromising patient safety, accelerating premature institutionalization, and overwhelming caregivers. Brexpiprazole marks a pivotal evolution in atypical antipsychotic therapy, providing a receptor-targeted, evidence-based intervention that addresses core monoaminergic dysregulation while avoiding the heavy sedation and motor impairment characteristic of off-label agents.

Keywords: Agitation, Alzheimer's Disease, Atypical antipsychotics, Brexpiprazole, Cohen-Mansfield Agitation Inventory (CMAI), International Psychogeriatric Association (IPA), Neuropsychiatric Symptoms, Serotonin-Dopamine Activity Modulator

EFFECTIVENESS OF A SELF-CARE COPING STRATEGIES INTERVENTION TO REDUCE DISTRESS AMONG DEMENTIA CAREGIVERS: A FEASIBILITY STUDY

Jeevasulochana Sinniah, Siti Anom Ahmad, & Halimatus Sakdiah Minhat

Abstract

Background: Informal caregivers of people living with dementia (PLWD), particularly those with behavioral and psychological symptoms of dementia (BPSD), often experience substantial emotional distress and caregiver burden. Evidence for individualised, culturally tailored psychoeducational interventions on self-care and coping strategies remains limited. Supporting caregivers' well-being may enhance the safe management of behavioral symptoms in PLWD, potentially reducing emotional distress, anxiety, depression and burden. This study evaluated the feasibility and preliminary effectiveness of the Self-CARE Coping Strategies Intervention in improving coping strategies and reducing caregiver distress among informal dementia caregivers. **Methods:** An explanatory research design was employed to evaluate the feasibility and preliminary outcomes of the Self-CARE Coping Strategies Intervention. Thirty informal caregivers participated in an 8-week individualised Self-CARE intervention via Google meet. The Self-CARE Coping Strategies Intervention was developed based on caregivers' needs and managing caregiving challenges. The intervention consists of four domains: Self-Calming, Self-Adaptive, Self-Resilient, and Self-Engagement. The Brief COPE and caregivers' distress related to BPSD (NPI-Q) questionnaire were assessed before and after the intervention, and open-ended question feedback explored participants' experiences and acceptability of the intervention. **Results:** Thirty caregivers completed the intervention. Most participants (83.4%) regularly practised the self-care strategies, indicating good engagement and acceptability. Participants also reported improved self-care, greater confidence in managing caregiving challenges, and increased willingness to seek support. Although overall coping scores did not significantly improve, caregiver distress associated with agitation and anxiety significantly decreased (both $p = 0.030$). **Conclusions:** The Self-CARE intervention was feasible and well accepted by informal dementia caregivers. Although improvements in overall coping strategies were not statistically significant, the intervention reduced caregiver distress associated with agitation and anxiety and showed promise as a culturally relevant psychoeducational intervention (Self-CARE Coping Strategies Intervention). A larger randomised controlled trial is warranted.

Keywords: Self-Care, Coping Strategies, Distress, Dementia, Caregiver.

LONGITUDINAL TRANSITIONS IN SOCIAL FRAILTY STATES AMONG COMMUNITY-DWELLING OLDER ADULTS IN MALAYSIA

Nur Aisyah Abdul Rahim, Choo Wan Yuen, Noran Naqiah Mohd Hairi, Halimah Awang, & Norma Mansor

Abstract

Social frailty is the loss, or risk of losing, resources needed to fulfil basic social needs. Despite its importance for healthy ageing, longitudinal evidence on changes in social frailty remains limited. Therefore, this study aimed to examine transitions in social frailty states among community-dwelling older adults. A two-year longitudinal study of 1,221 community-dwelling older adults aged ≥60 years was conducted using data from the Malaysia Ageing and Retirement Survey (MARS). Social frailty was assessed using the validated 15-item Malaysian Social Frailty Index (MySFI) and classified as social non-frailty, pre-frailty, or frailty. Transitions between social frailty states were categorised as unchanged, worsened, or improved. The overall distribution of social frailty states remained stable over two years ($p = 0.391$). However, substantial individual-level transitions were observed: 22.2% worsened, 20.9% improved, and 57.0% remained unchanged. Single-level transitions were most common, particularly from non-frailty to pre-frailty (42.4%) and from frailty to pre-frailty (42.8%). Two-level transitions between non-frailty and frailty were also observed (2.7% worsened; 4.9% improved), demonstrating the dynamic and bidirectional nature of social frailty. Social frailty states are dynamic among community-dwelling older adults, with a substantial proportion experiencing bidirectional changes over two years. Longitudinal monitoring may facilitate early identification of worsening social frailty and support timely interventions to promote healthy ageing.

Keywords: Social Frailty, Frailty Transitions, Longitudinal Study, Older Adults, Healthy Ageing

ASSOCIATION BETWEEN SOCIAL VULNERABILITY AND FRAILTY AMONG MIDDLE-AGED AND OLDER ADULTS IN MALAYSIA

Nur Azam Md Yusof, Noran Naqiah Mohd Hairi, Choo Wan Yuen, & Tan Chin Lung

Abstract

Background: Social vulnerability reflects the accumulation of social disadvantages that may reduce an individual's ability to anticipate, cope, resist and recover from an adverse event. However, its relationship with frailty among middle-aged and older adults in Malaysia remains underexplored. **Objective:** To investigate the association between social vulnerability and frailty among middle-aged and older adults in Malaysia. Methods: This secondary data analysis used data from the Malaysian Ageing and Retirement Survey (MARS). A preliminary SVI was constructed using the deficit accumulation approach, guided by a socio-ecological framework. Variables meeting predefined inclusion criteria were coded from 0 to 1 and aggregated into a composite index, with higher scores indicating greater social vulnerability. The distribution of SVI scores was described using mean and standard deviation. Spearman correlation analysis was used to assess the relationship between the SVI and the Frailty Index (FI). Multiple ordinal logistic regression was performed to examine the association between the SVI and frailty category, adjusting for age, sex and ethnicity. **Results:** A total of 5577 participants were included. The preliminary SVI comprised 31 social vulnerability deficits across domains including household composition, education, housing, health insurance, social connectedness, social networks, social participation, social support, wealth and living situation. SVI was normally distributed, with a mean of 0.43 (SD 0.12). SVI was positively correlated with FI ($\rho = 0.383$, $p < 0.001$). Every 0.1 unit increase in the SVI was associated with 59% higher odds of being in a higher frailty category (aOR 1.59; 95% CI 1.510-1.681; $p<0.001$) after adjusted for age, sex and ethnicity. **Conclusion:** Greater social vulnerability was associated with higher frailty among middle-aged and older adults in Malaysia. Further validation of the preliminary SVI is needed, including an assessment of its association with subsequent mortality.

Keywords: Social Vulnerability, Frailty, Population Ageing, Socio-Ecological Framework, Social Determinants of Health

BEYOND THE ELDERLY CARE CENTRE: DESIGNING ACTIVE AGEING ENVIRONMENTS FOR PARTICIPATION, CONNECTION AND CONTRIBUTION

Nor Zalina Harun, Dian Tri Irawaty, & Elisa Sutanudjaja

Abstract

Population ageing requires a fundamental shift in how community facilities for older people are conceived. This study explores how the emerging active-ageing agenda in Malaysia can be translated into a spatial design framework for a community-based care and activity centre. The study responds to a gap in existing age-friendly initiatives, where provision of health services, recreational activities, accessibility and care is increasingly evident, but the spatial integration of participation, productivity, intergenerational interaction and older people's agency remains less clearly articulated. The study adopts a qualitative exploratory approach combining literature and policy review with an inventory of active-ageing initiatives and field observations. The investigation draws on initiatives implemented by the Majlis Perbandaran Ampang Jaya (MPAJ), the Perak State Government and the Majlis Bandaraya Pulau Pinang (MBPP), including two community-garden initiatives, elderly activity programmes, care and dementia services, age-friendly parks, mobility support and ageing-in-place interventions. Findings from these initiatives were synthesised to identify recurring spatial needs and gaps across health and wellness, physical activity, social interaction, lifelong learning, productive ageing, intergenerational engagement, civic participation and connection with nature. The study proposes an Active Ageing Care and Activity Centre Framework comprising eight interconnected design components: care and wellness, active movement, social connection, productive ageing, learning and digital inclusion, intergenerational exchange, civic participation, and therapeutic and productive landscapes. Rather than treating these as discrete facilities, the framework integrates them through an active-ageing spatial spine that connects care, movement, social interaction, learning, creation and community contribution. The study argues that designing for active ageing requires a shift from providing services *for* older people towards creating environments *with and by* older people. The proposed framework offers a design-oriented approach for transforming care and activity centres into inclusive community hubs that support capability, independence, participation and meaningful contribution across later life.

Keywords: Elderly, Care Centre, Active Ageing, Environment, Participation

BRIDGING GENERATIONS: OLDER ADULTS' PERSPECTIVES ON AN INTERGENERATIONAL PROGRAM WITH YOUNG CHILDREN

Siti Noraini Asmuri, Iylia Dayana Shamsudin, & Halimatus Sakdiah Minhat

Abstract

Intergenerational programs provide opportunities for meaningful interaction between older adults and young children, with benefits for physical health, emotional well-being, and social connectedness. In Malaysia, structured intergenerational buddy programs involving older adults and young children remain underexplored, particularly from the perspectives of older adults. This study aimed to explore older adults' perspectives on intergenerational activities with young children. A qualitative study was conducted involving participants who were recruited based on being aged 60 years and above, able to communicate and understand Malay, and actively interacting with young children. Data were collected through a focus group discussion involving seven older adults aged 63 to 70 years, exploring participants' perspectives on the intergenerational program. The discussion was audio-recorded and transcribed verbatim. Qualitative content analysis (QCA) was utilized. Three themes emerged: (1) emotional fulfilment through interaction, (2) physical and mental health benefits, and (3) guidance through life experience. Emotional fulfillment through interaction reflects that positive relationships with younger generations may contribute to the feeling of happiness, connection, and social involvement. Shared activities such as reading, drawing, and cycling may encourage older adults to remain active, socially engaged, and mentally stimulated, hence improving physical and mental health status. Contributing to the younger generation by sharing knowledge, values, and practical life lessons highlights the meaningful role of older adults in guiding younger generations through life experiences. These findings suggest that older adults perceive intergenerational programs as meaningful platforms that can support health and well-being while providing opportunities to share knowledge, values, and life experiences with young children. It also highlights the potential value of incorporating older adults' perspectives into the development of intergenerational programs to ensure the activities are meaningful, culturally relevant, and according to their needs.

Keywords: Intergenerational Program, Perspectives, Older Adults, Young Children

FOSTERING OLDER ADULT WELL-BEING: QUALITATIVE INSIGHTS INTO INTERGENERATIONAL ACTIVITY DESIGN IN MALAYSIA

Iylia Dayana Shamsudin, Siti Noraini Asmuri, & Halimatus Sakdiah Minhat

Abstract

Intergenerational practice involving children and older adults is gaining increased attention due to its demonstrated benefits, including fostering positive attitudes and social competence among children, alongside improving mental health, physical well-being, and overall quality of life for older adults. However, research regarding intergenerational practice remains limited in Malaysia as the country transitions toward an aging nation. Therefore, this study aims to explore intergenerational activity design for promoting well-being from the perspectives of older adults. Seven older adults aged 60 to 70 years participated in a 120-minute focus group discussion (FGD). All participants were retired, actively interacted with children in their daily lives, and were recruited via call-for-participation flyers. The discussion was conducted in Malay, audio-recorded with written consent from all participants and transcribed verbatim for analysis. Iterative inductive qualitative content analysis revealed two primary themes regarding preferred intergenerational activity design: (1) moderately active activities, and (2) safety and health assurance. The findings suggest that moderately active engagement including light physical activities such as brisk walking, community improvement projects, and cooking is strongly recommended. Participants proposed that activities involving light movement are most beneficial, accommodating older adults' physical limitations while supporting their need for physical activity alongside children's energy. Furthermore, participants emphasized designing activities that are mutually enjoyable while guaranteeing safety and health. Shared concerns included amenity accessibility, physical environment, activity timing and duration, and children's age groups. These factors must be prioritized to create meaningful intergenerational engagement. This study highlights older adults' readiness for intergenerational programs when activities are meaningfully designed to balance both older and younger individuals' physical, social, health, and safety needs.

Keywords: Intergenerational Program, Activity Design, Older Adults, Young Children, Well-Being

FUNCTIONAL IMPAIRMENT IN LATER LIFE: MULTIDIMENSIONAL ASSOCIATIONS WITH NUTRITIONAL, ANTHROPOMETRIC AND LIFESTYLE FACTORS

Joanne Hui Xin Ang, & Yoke Mun Chan

Abstract

Frailty is a multidimensional geriatric syndrome characterised by reduced physiological reserve and increased vulnerability to adverse health outcomes. Weakness, slowness and low physical activity represent important physical manifestations of frailty that may compromise independence in later life. However, the relative contributions of nutritional, body composition, lifestyle and health-related factors to functional impairments remain incompletely understood. This cross-sectional study included 403 community-dwelling older adults. Functional impairment was defined as the number of impaired functional components of the Fried frailty phenotype (weakness, slowness, and low physical activity), and categorised as 0,1, ≥2 impairments, with higher scores indicating greater functional impairment. Ordinal logistic regression was performed to examine the independent associations of sociodemographic, anthropometric, nutritional, lifestyle and health-related factors with increasing functional impairment. Overall, 40.4% had no functional impairment, 36.5% had one functional impairment, and 23.1% had two or more functional impairments. Older age (OR=1.090, 95% CI:1.050-1.131, p<0.001), lower appendicular skeletal muscle mass index (ASMI) (OR=0.651, 95% CI:0.508-0.833, p<0.001), and higher waist-to-height-ratio (WHtR) (OR=1.925 per 0.1-unit increase, 95% CI:1.231-3.013, p=0.004) were independently associated with greater odds of functional impairment. For instance, each one-year increases in age was associated with a 9.0% increase in the odds of being a higher functional-impairment category, while each 0.1-unit increase in WHtR was associated with 1.93-fold higher odds. Chinese participants had lower odds of greater functional impairment compared with Indian participants (OR=0.462, 95% CI: 0.233-0.914, p=0.026). Greater functional impairment among community-dwelling older adults was independently associated with older age, lower skeletal muscle mass and greater central adiposity. In the context of rapid population ageing, these findings highlight the potential importance of muscle health and body composition in maintaining physical function in later life. Further investigation is warranted to establish the temporal relationships and identify predictors of subsequence functional deterioration.

Keywords: Frailty, Physical function, Multidimensional Association, Anthropometric, Nutritional

MEDICALLY TAILORED MEAL DELIVERY INTERVENTION IN VULNERABLE FOOD INSECURITY ELDERLY PATIENTS ON HAEMODIALYSIS IN KLANG VALLEY, MALAYSIA: A FEASIBILITY PILOT STUDY

Bee Swen Yeen, Chan Yoke Mun, Juju Nakasha binti Jaafar, & Yang Wai Yew

Abstract

Introduction: Food-insecure older haemodialysis patients are highly vulnerable to malnutrition. However, existing nutrition programme in Malaysia inadequately address their complex needs. This pilot study aimed to evaluate the feasibility and acceptability of medically tailored meal (MTM) programme delivered through an integrated farm-to-patient meal delivery model. **Methods**: A single-arm feeding trial was conducted in which participants received a four-week MTM intervention. The meals were tailored to renal nutritional requirements, with low sodium, potassium and phosphate content while providing high energy and protein. Meals were prepared in a commercially certified kitchen using vegetables sourced from a quality-certified farm to ensure food safety. The meals were delivered directly to participants during their regular haemodialysis sessions. Primary outcomes included recruitment, retention, adherence and acceptability, while secondary outcomes included food security and nutritional status. **Findings**: A total of 21 participants completed the intervention, with a mean age of 70.2 (SD=7.02). The intervention achieved a retention rate of 72.4% and 76.2% participants indicated that they would be willing to continue receiving MTMs, demonstrating good acceptability. The meal quality domain showed acceptable internal consistency (Cronbach's Alpha = 0.72). Out of total score of 4, mean satisfaction scores for meal attributes ranged from 3.0 to 3.9. Despite no statically significant changes in food security of protein-energy wasting (PEW) were observed, the prevalence of food insecurity decreased from 60.0% to 38.1%, while prevalence of PEW decreases from 21.1% to 4.8%. Statically significant changes were observed in serum potassium (p=0.03), pre-dialysis systolic blood pressure (p=0.02), and post-dialysis systolic blood pressure (p=0.04). These findings suggest that low sodium and low potassium MTMs help to improve serum potassium level and blood pressure control. **Conclusion**: This study shows potential as a scalable approach to support nutritional care for vulnerable elderly patients on haemodialysis and inform future large-scale clinical trials evaluating its effectiveness.

Keywords: Meal Delivery, Intervention, Vulnerable Food, Elderly, Haemodialysis

COMPARATIVE PERFORMANCE OF WAIST CIRCUMFERENCE AND BODY MASS INDEX FOR SARCOPENIC OBESITY SCREENING AMONG MALAYSIAN OLDER ADULTS WITH HYPERTENSION

Kai Xin Lu, & Yoke Mun Chan

Abstract

Background: Sarcopenic obesity (SO), characterised by the coexistence of reduced skeletal muscle mass and function with excess adiposity, is associated with adverse health outcomes including frailty and cardiovascular diseases. Older adults with hypertension are at higher SO risk, yet practical screening approaches remain limited. This study compared the discriminatory performance of anthropometric indicators for SO screening among Malaysian older adults with hypertension. **Methods & materials:** A total of 112 older adults (aged 60-84) were recruited from community and primary care settings. SO was defined using the Asia-Oceania consensus criteria based on body fat percentage, appendicular skeletal muscle index measured using bioelectrical impedance analysis, handgrip strength, and 5-times sit-to-stand test performance. Receiver operating characteristic (ROC) analysis evaluated the discriminatory ability of waist circumference (WC), body mass index (BMI), and calf circumference (CC). DeLong's test compared ROC curves (AUCs). **Results**: Among males, WC demonstrated acceptable discriminatory ability for SO screening (AUC=0.788, 95%CI=0.652, 0.925, p<0.001), performing numerically better than BMI (AUC=0.752, 95%CI: 0.586, 0.918, p=0.003), although the difference was not statistically significant (p=0.556). Adding CC to WC improved performance (AUC=0.847, 95%CI: 0.713, 0.980, p<0.001) but not significantly compared with WC alone (p=0.274). Adding BMI to the WC+CC model did not meaningfully enhance discrimination. Among females, WC demonstrated modest discriminatory ability (AUC=0.636, 95%CI: 0.449, 0.824, p=0.155), while BMI (AUC=0.552, 95%CI: 0.381, 0.723, p=0.552) and WC+CC (AUC=0.644, 95%CI: 0.459, 0.829, p=0.126) showed similarly limited performance. None of the anthropometric indicators achieved statistically significant discrimination for screening. **Conclusion:** Given the higher risk of SO among older adults with hypertension, BMI alone may be insufficient for screening. WC demonstrated promising discriminatory ability, particularly among males, highlighting the importance of central adiposity assessment in identifying SO risk. Incorporating accurate and standardised WC measurement into primary care and community practice may improve early SO detection among hypertensive older adults.

Keywords: Sarcopenic Obesity, Screening, Waist Circumference, Body Mass Index

MODALITY-SPECIFIC EFFECTS OF EARLY DARK-PHASE EXERCISE ON MUSCULAR AND BEHAVIOURAL OUTCOMES IN RATS WITH ALCL₃-INDUCED NEURODEGENERATION

Arimi Fitri Mat Ludin, Muhammad Hafiz Zuhdi Fairof, Amirul Hafiz Ahmad Abdullah, Nur Aisyah Anuar, Theng Choon Ooi, Farah Wahida Ibrahim, & Nor Fadilah Rajab

Abstract

Neurodegeneration is associated with deterioration in skeletal muscle mass, strength, locomotor function and memory. Exercise may attenuate these impairments, although its effects may differ according to exercise modality. This study compared the effects of aerobic, resistance and combined exercise performed during the early dark phase on muscular and behavioural outcomes in rats with aluminium chloride ($AlCl_3$)-induced neurodegeneration. Rats were assigned to five groups (n = 4 per group): non-induced sedentary control, $AlCl_3$-induced sedentary, $AlCl_3$ plus aerobic exercise, $AlCl_3$ plus resistance exercise, and $AlCl_3$ plus combined exercise. The study lasted eight weeks, with exercise performed during weeks 5–8. Gastrocnemius and soleus muscle masses were analysed using one-way analysis of variance. Grip strength, novel object recognition test (NORT) performance and open-field test (OFT) performance were evaluated longitudinally using mixed analysis of variance. $AlCl_3$ exposure significantly reduced grip strength ($p = 0.0315$), NORT performance ($p = 0.0016$) and OFT performance ($p = 0.0132$) from baseline to week 4. At week 8, the resistance-exercise group had greater gastrocnemius mass than the $AlCl_3$-induced sedentary group ($p = 0.0106$). Soleus mass was greater following aerobic and resistance exercise than in the $AlCl_3$-induced sedentary group (both $p = 0.0155$). From weeks 4–8, grip strength improved within the aerobic and resistance groups (both $p < 0.0001$). NORT performance improved within the resistance ($p = 0.0332$) and combined-exercise groups ($p = 0.0333$), while OFT performance improved within the aerobic ($p = 0.0039$) and combined-exercise groups ($p < 0.0001$). Early dark-phase exercise produced modality-specific muscular and behavioural benefits in rats with $AlCl_3$-induced neurodegeneration. Resistance exercise was associated with greater muscle mass and improved strength, while resistance and combined exercise improved recognition memory. Aerobic and combined exercise improved open-field performance. These preliminary findings support exercise as a potential strategy for addressing physical and behavioural consequences of neurodegeneration.

Keywords: Aluminium Chloride, Exercise Modality, Early Dark Phase, Muscle Mass, Grip Strength, Recognition Memory

IN VIVO GLYCEMIC SUPPRESSION BY PORANG GLUCOMANNAN (AMORPHOHALLUS MUELLERI) STARCH MATRICES: ESTABLISHING FOUNDATION FOR LOW-GI DYSPHAGIA BIGELS)

Amira Hasnanuha Ulayya, & Norhasnida Zawawi

Abstract

Background and objective: Aging populations frequently face the dual burden of dysphagia and complex co-morbidities, including age-related dysglycemia (e.g., type 2 diabetes) and pill-swallowing difficulties. Bigels, a combined system of hydrogel and oleogel, offer a transformative gerontechnological strategy as dysphagia food. However, starch-based hydrogels may elicit high glycemic responses, limiting their utility in diabetic elderly. Porang glucomannan (PGM) reportedly slows starch digestion by forming physical barrier over gelatinized starch, but the optimal concentration is undefined. This study determined the optimal PGM concentration in cooked cornstarch to minimize postprandial glucose response using a meal trained murine model. **Methods:** Male ICR mice (n=32) were meal trained and assigned to four groups (n=8): cooked cornstarch control (CC) and cooked cornstarch supplemented with 5%, 10%, or 15% PGM (w/w) (CGM 5%, CGM 10%, and CGM 15%). Baseline glucose tolerance was validated via an oral glucose tolerance test (OGTT). Postprandial blood glucose was measured at 0, 15, 30, 45, 60, 90, and 120 minutes post-ingestion to calculate incremental area under the curve (iAUC) and Glycemic Index (GI). **Result:** Pre-test OGTT confirmed uniform baseline glucose regulation ($p > 0.05$). Upon dietary challenge, all PGM-supplemented groups displayed flattened postprandial glucose curves. CGM 10% demonstrated optimal efficacy, yielding the smallest postprandial glucose increment and a significantly lower GI (6.33 ± 0.98) compared to the control diet (40.24 ± 9.43). Identifying 10% PGM as the optimal concentration for steric hindrance of alpha-amylase. **Conclusion and implications:** Porang glucomannan effectively blunts starch digestion in vivo, with 10% PGM serving as the optimal functional level. Hybridizing this optimized starch-PGM matrix with structured oleogel, potentially designed a bigel matrix to satisfy IDDSI Level 5 swallowing safety, not only for low-glycemic nutrition but also a versatile carrier for bioactives or oral medicines in elderly care.

Keywords: Porang's Glucomannan, Cornstarch, Dysphagia, Gels, Bigels, Low-Glycemic Index, Elderly Nutrition

RANDOMIZED CONTROLLED HUMAN TRIAL ON THE EFFECT OF EDIBLE BIRD NEST SUPPLEMENTATION FOR SKIN AND CARDIOVASCULAR PROFILE IN PREMENOPAUSAL WOMEN: A PILOT STUDY

Salwana Ahmad, Kim Wei Chan, Der-Jiun Ooi, Norsharina Ismail, Norazlinaliza Salim, & Md Zuki Abu Bakar

Abstract

Cardiometabolic and menopausal changes may begin before menopause in women, but Edible's bird nest (EBN) potential health benefits in this population are unclear. This study aimed to evaluate the effects of EBN supplementation on menopausal symptoms, abdominal ultrasound parameters, reproductive hormonal and inflammatory biomarkers, cardiovascular risk profiles, and skin ageing parameters among premenopausal women with borderline hypercholesterolemia. A 12-week randomised controlled trial involving 39 premenopausal women with borderline hypercholesterolemia was conducted to receive either 250 g/day of EBN mixture or a matched jelly. Thirty-nine participants (EBN, n = 21; placebo, n = 18) completed the 12-week study. Mean age was 44.76 ± 4.13 years in the EBN group and 43.89 ± 3.91 years in the placebo group ($p = 0.504$). Significant between-group differences were observed in post-intervention uterine volume ($p = 0.031$) and mean ovarian volume ($p = 0.038$). In the EBN group, uterine volume decreased significantly ($p = 0.019$), with reductions in IL-6 ($p = 0.026$) and TNF-α ($p = 0.040$). No significant changes were observed in menopausal symptoms, estradiol, progesterone, testosterone, LH, or FSH. Skin hydration increased significantly in the EBN group at the cheek ($p = 0.002$), backhand ($p = 0.002$), forearm ($p = 0.001$), and leg ($p = 0.033$), with significant between-group differences at the backhand ($p = 0.015$) and forearm ($p = 0.004$). Significant improvements in elasticity were observed across several sites for R2, R5, and R7. A significant time effect was observed for LDL cholesterol ($p = 0.022$). Apolipoprotein C-II ($p = 0.047$) and C-III ($p = 0.001$) differed significantly between groups, with significant post-intervention differences for both Apo C-II and Apo C-III (both $p = 0.001$). EBN supplementation showed beneficial effects on selected cardiovascular and skin health parameters, supporting women's health, early risk reduction and quality of life during the menopausal transition.

Keywords: Edible Bird’s Nest, Premenopausal Women, Borderline Hypercholesterolemia, Cardiovascular Risk Factors, Skin Ageing, Reproductive Hormones, Inflammatory Biomarkers

INSIGHTS INTO DIGITAL MENTAL HEALTH READINESS AMONG EMOTIONALLY DISTRESSED OLDER ADULTS IN KELANTAN: A PRELIMINARY FINDING

Nurul Afiqah Che Mohd Zaid, Ahmad Shahril Ab. Halim, Wan Nor Arifin Wan Mansor, & Asrenee Ab Razak

Abstract

Background: Limited health literacy among elderly, as reported in the 2023 Malaysia Health Literacy Survey (MHLS), coupled with increasing digitalization of healthcare, underscores the need to investigate digital mental health literacy (DMHL) among emotionally distressed elderly requiring timely mental healthcare. However, evidence on DMHL and attitudes towards professional mental healthcare remains scarce. **Objective:** To explore DMHL and mental health help-seeking attitudes (MHSA) among emotionally distressed older adults in Kelantan, as preliminary insights into readiness for digital mental healthcare. **Methods:** A cross-sectional analytical study was conducted among 162 individuals aged ≥60 years attending Family Medicine and Psychiatric Outpatient Clinics at Hospital Pakar Universiti Sains Malaysia (HPUSM), Kelantan. Participants with GDS-15 and/or GAD-7 scores of ≥5 were included, while technology-illiterate elderly were excluded. DMHL was evaluated using a modified eight-item Malay eHealth Literacy Scale (eHEALS) contextualised to mental health, while MHSA were assessed using nine-item Malay Mental Health Help-Seeking Attitudes Scale (MHSAS). Descriptive statistics were used. **Results:** Mean age was 67.73 ± 5.78 years. Mean DMHL was 27.59 ± 7.60 out of 40, indicating moderate level of DMHL. Participants reported greatest confidence in distinguishing good- from poor-quality online mental health resources (3.93 ± 1.21), but were least confident in using internet to answer mental health-related questions (3.23 ± 1.04). MHSA was highly favourable, with mean MHSAS 6.54 ± 0.59 out of 7, with 40.7% achieved maximum score of 7.00. Professional mental healthcare was perceived as important (6.68 ± 0.58) and useful (6.64 ± 0.69), while healing (6.41 ± 0.82) and effectiveness (6.46 ± 0.80) received lower ratings. **Conclusion:** These findings suggest potential digital mental health readiness among emotionally distressed elderly. Despite highly favourable attitudes towards professional mental healthcare, lower ratings for healing and effectiveness may reflect residual stigma. Digital mental health thus offer an avenue to professional support for elderly.

Keywords: Digital, Mental Health, Emotional, Distress, Older,

PSYCHOSOCIAL RISK FACTORS FOR DEPRESSION IN THE POST-RETIREMENT PERIOD: A CROSS-SECTIONAL STUDY OF MALAYSIAN GOVERNMENT PENSIONERS

Ahmad Shahril Ab. Halim, Raishan Shafini Bakar, & Sharifah Shakirah As-Sahab Wan Muhamad Nasir

Abstract

Background: The transition to retirement represents a significant life change that can impact the mental health of older adults, influenced by various psychosocial factors. Malaysia's ageing population and expanding pool of government pensioners make this transition increasingly relevant to public health planning, yet local data on depression and loneliness among pensioners remain limited. **Objective:** This study investigated the prevalence of depression and loneliness and identified associated factors among government pensioners in Kelantan, Malaysia. **Methods:** A cross-sectional study was conducted from March 2022 to March 2023, involving 378 members of the Malaysian Government Pensioners Association (Kelantan branch), recruited through the association's membership database. Data were collected via an online questionnaire using the Malay version of the Geriatric Depression Scale-14 (M-GDS-14) and the De Jong Gierveld Loneliness Scale, along with sociodemographic variables, including living arrangements and perceived social support. Logistic regression was used to identify independent predictors of depression, adjusting for potential confounders. **Results:** The mean age of participants was 65.7 (SD 3.60) years, with 61% being male. The prevalence of depressive symptoms and loneliness was 13.2% and 16.1%, respectively. Regression analysis revealed three factors independently associated with depression: loneliness (Adjusted Odds Ratio [aOR] = 2.57, 95% CI 1.23 to 5.36, *P*=0.012), living alone (aOR = 2.38, 95% CI 1.03 to 5.49, P=0.045), and poor perceived social support (aOR = 3.11, 95% CI 1.37 to 7.07, P=0.007). **Discussion:** Depression prevalence in this cohort was relatively low compared with other older adult populations, though loneliness and social isolation emerged as consistent, significant risk factors. This suggests that even where depressive symptoms are uncommon, the social dimensions of retirement, particularly living arrangements and perceived support, warrant closer attention. These findings support targeted interventions to strengthen social engagement and support networks for pensioners, pointing to social connectedness as a modifiable target for mental health promotion**.**

Keywords: Depression, Anxiety, Loneliness, Elderly, Older Adults, Pensioners, Mental Health, Ageing

SUSTAINING GROUP SINGING IN DEMENTIA CARE: FACILITATION, ENVIRONMENT AND PERSON-CENTRED ADAPTATION IN A LONG-TERM QUALITATIVE DUAL-CASE STUDY IN GERMANY

Tan Teik Poi

Abstract

Group singing is widely promoted in dementia care, yet much of the literature focuses on short-term interventions, with less attention to how singing groups are organised, adapted and sustained in everyday residential practice. This qualitative dual-case study examined group singing in two dementia-specialised residential care centres (RCCs) in North Rhine-Westphalia, Germany, through prolonged engagement of 112 weeks at RCC1 and 96 weeks at RCC2 (October 2022–December 2024). Thirty residents with mild-to-moderate dementia (7 men, 23 women; mean age 86.2 years, range 78–100) participated in regular sessions. Data comprised semi-structured interviews and informal conversations with ten caregivers and six family members, together with observational field notes and reflexive memos generated by the researcher-facilitator, and were analysed thematically (Braun & Clarke, 2006). Four higher-order themes were developed. Caregivers and family members perceived group singing as supporting confidence, mood and social connection, particularly among residents who were otherwise less socially engaged. Positive responses were not universal: some residents experienced overstimulation or disengagement, highlighting the importance of respecting preferences and offering person-centred alternatives rather than expecting participation. Sustained engagement was shaped by environmental and organisational conditions, including room and group size, scheduling and transport logistics, as well as facilitation combining predictable structure with individualised musical and vocal-pedagogical adaptation. Non-verbal responses also provided important indications of engagement among residents with limited verbal communication. The study's most distinctive contribution lies not in confirming that singing is enjoyable, a well-established finding, but in describing the conditions and facilitation practices that shape whether group singing programmes are sustained meaningfully in residential dementia care, including for residents for whom singing is not appropriate.

Keywords: Dementia Care, Group Singing, Person-Centred Care, Facilitation, Social Participation, Qualitative Case Study

MAPPING QUALITY OF LIFE AND ASSOCIATED FACTORS AMONG COMMUNITY-DWELLING OLDER PERSONS USING SPATIAL AND NON-SPATIAL DATA IN SEPANG, SELANGOR: A STUDY PROTOCOL

Syaza Atikah Saidi Mohammad, Nor Afiah Mohd. Zulkefli, & Siti Aisah Mokhtar

Abstract

Background: Malaysia is undergoing a demographic transition towards an ageing population, posing significant implications for age-friendly planning. Quality of life (QoL) in later life is shaped not only by individual and social circumstances but also by the surrounding neighbourhood environment, yet local evidence remains limited and relies largely on perceived rather than objectively measured environmental exposure. This study aims to assess QoL and identify associated individual, social and Geographic Information System (GIS)-derived built environment factors among community-dwelling older persons in Sepang, Selangor. **Methods:** An analytical cross-sectional study will be conducted among 370 community-dwelling older persons aged 60 years and above in Sepang, Selangor. Cluster sampling will be applied, with selected Klinik Kesihatan in each mukim (Dengkil, Labu and Sepang) serving as clusters, followed by systematic sampling of eligible attendees at the general outpatient (cold-case) clinic. Data will be collected using a validated self-administered questionnaire. Statistical and spatial analyses will use IBM SPSS and QGIS, respectively. QoL will be the dependent variable, measured using the 13-item WHOQOL-AGE and classified as good versus poor. Independent variables will include sociodemographic characteristics, physical function, depression, comorbidity, lifestyle, social support, social participation and GIS-derived built environment factors obtained from open-source spatial datasets, namely residential density, accessibility to community and healthcare services, walkability, land use mix and green spaces. Bivariate analysis will be performed using simple logistic regression, followed by multiple logistic regression, a two-part availability-intensity model for zero-heavy exposures, multiscale buffer comparison and CHAID decision-tree analysis. **Expected outcomes:** The study will describe QoL and its spatial distribution, identify individual, social and environmental determinants, and determine the neighbourhood scale at which built environment exposure is most strongly associated with QoL. **Conclusion:** Findings will provide district-level spatial evidence to guide age-friendly planning and targeted interventions for older persons in Malaysia.

Keywords: Quality of Life, Older Persons, Geographic Information System, Built Environment, Healthy Ageing, Spatial Data, Age-Friendly Environment

INTEGRATING AGEING-IN-PLACE PRINCIPLES INTO MALAYSIA'S HOUSING DEVELOPMENT

Lina Emira Ahmad Adni, & Nor Zalina Harun

Abstract

Population ageing is reshaping housing needs and presents an increasing challenge for the planning and development of supportive living environments in Malaysia. As Malaysia moving towards ageing nation, housing must respond to changing needs while enabling elderly to maintain independence and remain within their homes and communities. Ageing-in-place therefore requires consideration not only within ageing policy but also within mainstream housing and built-environment planning. Malaysia has established policy and planning provisions that recognize older persons and support age-friendly living environments within the household surroundings. However, preliminary review suggests a gap between ageing and housing policy. While ageing policy emphasizes independence and supportive environments, these aspirations appear less explicitly translated into mainstream housing policy and development guidance. This raises questions about the extent to which ageing-in-place principles are integrated across Malaysia's policy landscape. This study aims to examine the integration of ageing-in-place principles across Malaysia's ageing and housing policy frameworks and identify opportunities to strengthen their translation into mainstream housing development. A qualitative policy analysis will be conducted through document review and comparative policy mapping of five key national instruments: the National Policy for Older Persons (DWEN), Second National Urbanisation Policy (DPN2), Fourth National Physical Plan (RFN4), National Housing Policy (DRN), and Housing Planning Guidelines (GPP Perumahan). The analysis will identify policy alignment, gaps, and the extent to which provisions supporting elderly and ageing-in-place are incorporated across these instruments. The study is expected to identify areas of policy convergence and fragmentation and recommend approaches for embedding ageing-in-place principles within mainstream housing planning and development. Theoretically, it advances ageing-in-place through a policy-integration perspective, while practically informing more coordinated and age-responsive housing development in Malaysia.

Keywords: Ageing in Place, Age-Friendly Housing, Older Adults, Policy Integration, Housing Planning

BRIDGING CARE NEEDS AND HOME ADAPTATION: HOME MODIFICATION AMONG FAMILY CAREGIVERS OF OLDER ADULTS IN MALAYSIA

Rahimah Ibrahim, Norisma Aiza Ismail, Mohd Fadzillah Bagat, Mohd Rizal Hussain, Ajun Chin, Siti Anom Ahmad, Sarifah Azizah Haron, Asmidawati Ashari, Shahrul Bahyah Kamaruzzaman, Roshanim Koris, & V Santha Vaithilingam

Abstract

Background: Ageing in place is a key priority in Malaysia's ageing policy. However, accessibility standards do not extend to private dwellings, and no national programme provides dedicated support for age-related home modifications. Consequently, home adaptation largely falls on families, yet its extent, costs, and relationship with care needs and household resources remain poorly documented. Objectives: To determine the proportion of family caregivers who had modified their homes, the expenditure incurred, and the association of home modification with care recipient needs and household income. Methods: A cross-sectional survey was conducted among 426 family caregivers recruited through purposive sampling. Home modification data were available for 371 caregivers. Modification was assessed through caregiver-reported modification status (yes/no), with expenditure recorded in Malaysian Ringgit (RM). Care needs were assessed across seven activities requiring supervision, guidance, or assistance. Differences by modification status were examined using independent-samples t-tests. Binary logistic regression assessed factors associated with home modification, including caregiver age, sex, marital status, education, household size, care needs, and household income. Results: Overall, 302 (70.9%) caregivers reported that the older adults required at least one form of supervision, guidance, or assistance. Among 371 caregivers with valid modification data, 110 (29.6%) reported modifying their homes, with a median expenditure of RM5,000 (IQR: RM100–RM200,000). Among caregivers who modified their homes, 97 (88.2%) cared for older adults requiring supervision, guidance, or assistance. However, 115 care recipients requiring assistance with more than three activities lived in unmodified homes. Care needs did not differ significantly by modification status, $t(296) = -1.208$, $p = .200$, nor did household income, $t(365) = -1.354$, $p = .669$. The multivariable logistic regression model was not statistically significant, $\chi^2(16, N = 259) = 17.913$, $p = .329$, with Cox and Snell $R^2 = .067$ and Nagelkerke $R^2 = .093$. Care needs (OR = 1.078, 95% CI: 0.954–1.218, $p = .224$) and household income (overall $p = .938$) were not significantly associated with home modification. Conclusion: Home modification was uncommon despite substantial care needs. The presence of care recipients with higher assistance needs in unmodified homes suggests a potential gap between care requirements and environmental adaptation. Routine home assessment, occupational therapy referral, caregiver education, and targeted housing support may facilitate safer ageing in place.

Keywords: Ageing in Place, Family Caregivers, Home Modification, Older Adults, Housing Adaptation

BRIDGING THE GAP IN GERIATRIC CARE: A MULTICENTER EVALUATION OF PSYCHOSOCIAL INEQUITIES AND CAREGIVER BURDEN IN MALAYSIAN TERTIARY HOSPITALS

Sivasangari Subramaniam, Thiam Chiann Ni, Mogana Thulaisi, Norshahida Binti Abdul Hamid, Keng Jun Ni, Ang Zi Heng, Adilah Aminuddin, Sai Deepikaa Sotheenathan, & Sotheenathan Krishinan

Abstract

Background: As Malaysia's aging demographic expands, family caregivers serve as the foundational pillar of long-term care, absorbs the complex physical and psychosocial demands of older adults. Within social gerontology, caregiving operates as a critical structural determinant of health; yet, institutional support for these families remains profoundly underprioritized in routine clinical workflows. Understanding the intersection of caregiver strain, cultural barriers, and systemic gaps is vital to cultivating equitable, sustainable solutions for an aging society. Objectives: This multicenter study aimed to quantify caregiver burden and unmet mental health needs among families of elderly outpatients in Malaysian tertiary hospitals, while identifying the distinct sociocultural and structural barriers impeding access to support services. Methods: A cross-sectional design was deployed across outpatient clinics at selected tertiary medical centers. Data collection leveraged a three-tier framework: structural clinical audits of patient health profiles, the 12-item Zarit Burden Interview (ZBI-12) evaluating subjective caregiver strain, and a customized inventory mapping perceived barriers to psychosocial care. Results: Preliminary analyses revealed that over 65% of family caregivers experience mild-to-high levels of burden, characterized by profound emotional exhaustion and an acute deficit in social respite. Sociocultural stigma, deficient resource awareness, and practical time-cost constraints emerged as primary structural barriers to seeking help. Crucially, caregivers managing older adults with progressive cognitive decline or active psychiatric comorbidities exhibited significantly elevated ZBI-12 scores, exposing an exceptionally vulnerable cohort facing severe social isolation. Conclusion: Caregiver distress represents a critical social inequity that threatens the viability of family-centered eldercare in Malaysia. These insights underscore the urgency of integrating routine caregiver vulnerability screenings and targeted community respite programs into national health policies, thereby preserving the structural well-being of both older adults and their support systems.

Keywords: Social Gerontology, Caregiver Burden, Psychosocial Strain, Stigma, Zarit Burden Interview, Malaysia

DEVELOPING THE SUCCESSFUL AGEING READINESS FRAMEWORK (SARF): A MIDLIFE APPROACH TO HEALTHY AND SUCCESSFUL AGEING IN MALAYSIA

Yeshulavani Rajeswaran, Mohammad Mujaheed Hassan, Azlina Mohd Khir, & Asmidawati Ashari

Abstract

Introduction: Population ageing has intensified global interest in promoting healthy, active, and successful ageing. However, existing ageing frameworks predominantly focus on later-life outcomes or interventions targeting older adults, with limited attention given to preparedness for ageing during middle adulthood. Despite growing evidence that many determinants of later-life wellbeing are established during midlife, there remains no integrated framework specifically designed to conceptualise readiness for successful ageing. **Methods:** This study employed a narrative literature review and conceptual framework development approach. Literature published between 2000 and 2026 was reviewed across major databases, including Scopus, Web of Science, PubMed, and Google Scholar. Key theories, ageing models, and determinants associated with positive ageing outcomes were synthesised to identify core dimensions of ageing preparedness. **Results:** The review led to the development of the Successful Ageing Readiness Framework (SARF), a multidimensional framework comprising five interrelated domains: Psychological Readiness, Health Readiness, Social Readiness, Financial and Occupational Readiness, and Purpose and Developmental Readiness. Grounded in Life Course Theory, Selection, Optimisation and Compensation (SOC) Theory, and Human Ecology Theory, the framework proposes that these domains collectively contribute to Successful Ageing Readiness, which subsequently influences later-life outcomes such as wellbeing, independence, life satisfaction, active participation, healthy ageing, and successful ageing. **Conclusion:** SARF introduces a readiness-based perspective that shifts the focus from evaluating ageing outcomes to assessing preparedness for successful ageing during middle adulthood. By positioning midlife as a critical intervention period, the framework advances a proactive and preventive approach to ageing research and practice. The framework offers potential applications in individual assessment, workplace wellbeing programmes, community interventions, and ageing policy development, while providing a foundation for future measurement tools and empirical validation studies.

Keywords: Successful Ageing Readiness Framework (SARF), Successful Ageing, Healthy Ageing, Middle Adulthood, Ageing Preparedness, Life Course Theory, Social Gerontology

SUB-DISTRICT MAPPING OF AGED POPULATIONS IN MALAYSIA: LESSONS AND POTENTIAL

Chai Sen Tyng, Danesh Prakash Chacko, Rahimah Ibrahim, Muhammad Mujaheed Hassan, & Abdul Rashid Mohamed Shariff

Abstract

Population ageing is spatially uneven, yet national and state-level statistics often conceal important differences between local areas. Understanding where older persons live, how their numbers are changing, and whether health and social services can meet their needs is essential for developing more equitable and responsive ageing policies. Sub-district mapping offers an opportunity to move beyond aggregate demographic indicators towards a more geographically informed understanding of population ageing and its implications for local planning. This paper examines the lessons and potential of sub-district mapping of aged populations in Malaysia, focusing on the impact of small-area demographic information for identifying local variations in ageing and informing policy and service delivery. It considers how census data, population estimates and geospatial analysis can be used to examine the distribution, concentration and changing composition of older populations across localities, especially when boundaries change. Such analysis can reveal patterns that may be obscured by district-level averages, including differences between urban and rural communities and localities experiencing different demographic trajectories. The paper further discusses the potential of integrating demographic mapping with socioeconomic indicators, health and welfare information, transport accessibility, and the geographical distribution of care services. This broader approach can help distinguish areas with high concentrations of older persons from those facing greater vulnerability, inadequate health coverage or emerging pressures on long-term care. However, the usefulness of such mapping depends on data availability, geographical comparability, temporal consistency and appropriate analytical interpretation of small-area estimates. Demographic maps alone cannot establish individual vulnerability or fully capture informal care networks and community resilience without additional data. This paper argues that sub-district mapping should be developed not merely as a descriptive exercise, but as a practical tool for place-based ageing policy with local governments. Strengthening data integration, inter-agency collaboration and local validation could support more targeted resource allocation, age-friendly community development and the planning of accessible health, social and long-term care services. Ultimately, mapping can help bridge the gap between demographic evidence and local action, supporting more equitable opportunities to age with dignity, security and participation throughout Malaysia.

Keywords: Population Ageing, Sub-District Mapping, Geospatial Analysis, Local Planning, Long-Term Care, Malaysia

ABSTRACT BOOK

02

Poster Presentation Abstracts

19–21 October 2026

KLGCC Convention Centre, Kuala Lumpur

POPULATION AGEING AND LONGEVITY IN MALAYSIA: TRENDS IN MORTALITY AND EXPECTED RETIREMENT DURATION

Halim Shukri Kamaruddin

Abstract

Malaysia is experiencing a gradual demographic transition characterised by population ageing and improvements in longevity, which has significant consequences for the duration of life spent beyond conventional retirement ages. This study examines long-term trends in population ageing, mortality, and expected retirement duration in Malaysia using annual population and mortality data from the Department of Statistics Malaysia (DOSM). The analysis focuses on changes in the older population and mortality patterns at later ages. Age-specific mortality rates are used to evaluate mortality improvements over time, while life-table techniques are applied to calculate remaining life expectancy at age 60. Expected retirement duration is defined as the expected remaining lifetime at age 60, in alignment with Malaysia's existing minimum retirement age. The analysis also investigates trends in the percentage of individuals aged 60 and older, along with specific indices of population ageing, to offer a more comprehensive demographic perspective. The research aims to illustrate how the reduction in mortality rates among the elderly has led to prolonged life expectancy past the age of 60 and an increasing expected duration of retirement. The research offers a demographic viewpoint on the evolving duration of later life in Malaysia by integrating population-ageing indicators with mortality rates and residual life expectancy. The findings are intended to contribute to discussions on population ageing, retirement readiness, and the implications of increasing longevity for individuals and society as Malaysia transitions towards an aged population.

Keywords: Population Ageing, Longevity, Mortality, Remaining Life Expectancy, Retirement Duration

NAVIGATING THE DEFICIT AND NOSTALGIA VIEW TO STRATEGIC VALUE OF AGEING WORKFORCE USING A FOUR-QUADRANT MODEL FOR SUSTAINABLE AI-HRM

Low Mei Peng, Har Wai Mun, & Ting Li Han

Abstract

Ageing workforces are often viewed through two counterproductive lenses, i.e., the Deficit-Based Trap, which views older workers through stereotypes of decline, low adaptability, and high cost; and the Nostalgia-Based Trap, which, while valuing their experience, risks framing them as artefacts of the past. Both perspectives limit opportunities for employability, inclusion, and meaningful contribution among the older workers. This conceptual paper challenges these traps and proposes a new framework for strategic, sustainable HRM enhanced by Artificial Intelligence (AI) technologies. Drawing on theories of social identity, human-AI collaboration, and sustainable HRM, we introduce a Four-Quadrant Model that reimagines the role of older workers. The model maps organizational mindsets with vertical axis: View of Older Workers (as a "Problem" vs. "Asset") and horizontal axis: Focus of HR Strategy (on "Efficiency/Cost-Cutting" vs. "Sustainable Value Creation"). By integrating AI-supported tools for personalized learning at individual level, bias-free recruitment, and job redesign at organizational level, organizations can transition from the "Age as Liability" quadrant to a future-oriented "Age as Strategic Advantage" quadrant. Additionally, we argue that AI, when governed ethically, can be a powerful tool to dismantle ageist assumptions, foster intergenerational collaboration, and align workforce management with the SDGs to foster decent work and reduce inequalities. It concludes with practical recommendations for managers and policymakers on building an age-inclusive, technologically empowered, and sustainable future for work.

Keywords: Ageing Workforces, AI-Enabled Sustainable HRM, Decent Work, SDG 8, SDG 10, Silver Economy

LOCALISING DIGITAL LITERACY TRAINING FOR OLDER PERSONS IN INDONESIA: A TRAINING NEEDS ANALYSIS THROUGH STAKEHOLDER FOCUS GROUP DISCUSSION

Siti Farra Zillah Abdullah, Mohamad Fazdillah Bagat, Siti Anom Ahmad, Farahdibha Tenrilemba, Fakhrul Zaman Rokhani, & Nur Ainani Mahyuddin

Abstract

Digital literacy is increasingly important for older persons to access information, healthcare, social connections, and digital services. However, digital literacy training may need to be adapted to local technologies, digital practices, accessibility needs, and online risks. This study conducted a Training Needs Analysis (TNA) to identify stakeholder-informed training and adaptation needs for localising the MyAgeing® Digital Literacy Modules to the Indonesian context. The TNA was conducted through Focus Group Discussion on 11 May 2026 in Jakarta, Indonesia, involving 31 purposively selected stakeholders from government (n=10), universities (n=8), and civil society/community organisations (n=13). Participants reviewed the modules for relevance, usability, contextual appropriateness, technology applicability, and digital safety. Data were analysed using thematic analysis. Five interrelated training needs emerged. First, accessibility required simplified language, larger fonts, and audio and visual support to accommodate diverse functional abilities. Second, instructional design required clearer step-by-step guidance, infographics, visual cues, and separate learner and facilitator manuals. Third, contextual relevance highlighted Indonesian examples, local telecommunications providers, nationally relevant health, navigation, communication, and public-service applications, and social media platforms. Fourth, technology-related needs reflected perceived gaps between existing content and current digital practices, particularly the need for Android-focused materials and updated content addressing emerging technologies and associated risks, including AI-enabled scams and fraud. Finally, digital safety emphasised scam awareness, misinformation verification, and practical online safety guidance. The TNA identified stakeholder-informed needs across accessibility, instructional design, contextual relevance, technology use, and digital safety. The findings indicate that localisation extends beyond linguistic adaptation to encompass technological, institutional, instructional, and socio-contextual dimensions. Addressing these needs could strengthen the effectiveness, usability, and contextual relevance of digital literacy training for older persons in Indonesia. The findings provide a basis for localising digital literacy modules to better support older persons' meaningful participation in an increasingly digital society.

Keywords: Digital Literacy, Older Persons, Training Needs Analysis, Programme Localisation, Focus Group Discussion

FORMAL EDUCATION AND LIFELONG LEARNING FOR HEALTHY AGEING IN SOUTH EAST ASIA

Maria Alaguru

Abstract

In response to the United Nations Decade of Healthy Ageing, this study provides a review of literature on the role and impact of education and lifelong learning on healthy ageing, to propose policy recommendations to education policymakers in South East Asia (SEA). The study is situated in a theoretical framework considering both conflict and consensus theories, building on positional conflict theory, functionalism and human capital theory. It critiques neoliberalism and globalisation forces in education and the job market, and relates learning theories to the learning society in a sociocultural perspective, towards enhancing education and informal learning to support healthy ageing. Education should take a life-course approach, ensuring equitable access to inclusive and quality education earlier rather than later in life. It should focus on teaching students to be self-directed lifelong learners. The study argues that government interventions in formal education and informal learning are vital for securing ageing societies, and to ensure that social institutions including educational institutions guard against market rules and discrimination. Public support is required for vulnerable groups, carers, community-based education providers and employers hiring older adults. To prevent a care crisis, regulations, infrastructure, resources and licensed manpower are needed for a sustainable, quality caregiving industry.

Keywords: Lifelong Learning, Lifelong Learner, Healthy Ageing, Educational Policy, Caregivers, Carers, Older Adult Workers, Educational Quality, Access to Education

FAMILY CAREGIVING FOR OLDER ADULTS: BALANCING CAREGIVING CHALLENGES AND POSITIVE EXPERIENCES

Norisma Aiza Ismail, & Rahimah Ibrahim

Abstract

Population ageing has increased demand for family-based care. While caregiving may foster fulfilment and strengthen family relationships, it can also impose physical, emotional, social, and financial demands. Evidence on how caregiving difficulties and positive experiences coexist among Malaysian family caregivers remains limited. This study examined perceived caregiving difficulty, positive caregiving experiences, and their association among family caregivers of older adults in Malaysia. This cross-sectional study used primary data from the Long-Term Research Grant Scheme (LRGS) project *Psychosocial and Societal Burden of Cognitive Frailty among Older Persons*, collected through purposive community sampling between 2022 and 2024. The sample comprised 426 family caregivers of older adults with mixed health conditions (mean age = 50.9 years, SD = 13.0; 76.3% female; 73.9% co-residing with care recipients). Perceived caregiving difficulty was assessed using a single 10-point item (1 = very easy, 10 = very difficult), while positive caregiving experiences were measured using the 9-item Short Positive Aspects of Caregiving Scale (S-PAC; α = .93). Descriptive statistics, Pearson correlation, and t-test were computed using SPSS. Mean perceived caregiving difficulty was 5.49 (SD = 2.44), indicating moderate difficulty. Positive caregiving experiences were relatively high (M = 4.04, SD = 0.71, n=368), with item means ranging from 3.83 to 4.30. The highest-rated item was that caregiving helped caregivers appreciate life more (M = 4.30), while the lowest was caregivers' sense of being valued (M = 3.82). Difficulty was weakly but significantly negatively correlated with positive caregiving experiences ($r = -.15$, $p = .006$). Co-residing caregivers reported significantly greater difficulty than those living apart ($t = 2.63$, $df = 351$, $p = .009$). Caregiving difficulties and positive experiences largely coexist rather than trade off. Support should therefore adopt a holistic approach combining practical skills, emotional support, coping strategies, and formal recognition, particularly for co-residing caregivers.

Keywords: Caregiving, Older, Challenges, Positive, Experiences

IN VITRO HEPATO-PROTECTIVE OF MIXED BERRIES POWDER, BET-AGING™

Che Muhammad Khairul Hisyam Bin Ismail, Mohammad Adi Bin Mohammad Fadzil, Suraiami Binti Mustar, Mohd Naeem Bin Mohd Nawi, Noor Farhana Binti Ab Karim, & Aswir Bin Abd Rashed

Abstract

Population ageing has emerged as a global demographic trend driven by declining birth rates and increasing life expectancy. Malaysia officially became an ageing nation in 2025, with individuals aged 65 years and above comprising 8% of the population, according to the United Nations definition. As population ageing increases the prevalence of chronic diseases and functional decline, there is a growing need for interventions that promote healthy ageing and reduce the associated economic burden. A key molecular hallmark of ageing is the progressive decline in intracellular nicotinamide adenine dinucleotide (NAD^+) levels, which impairs mitochondrial function, increases oxidative stress and DNA damage, and contributes to cellular dysfunction. Consequently, nicotinamide mononucleotide (NMN)-based supplementation has emerged as a promising strategy to restore NAD^+ levels and mitigate age-associated health deterioration. Bet-Aging™ (BA) is a commercially available NMN-based health supplement formulated with mixed berries extract, acerola cherry extract, and inulin, which may provide additional health benefits. In contrast, liposomal BA incorporates liposomal delivery technology to enhance the solubility of the formulation. This study aimed to evaluate the cytotoxicity of an NMN-based health supplement, BA using the human hepatocellular carcinoma cell line (HepG2). HepG2 cells were exposed to various physiologically relevant concentrations of conventional BA and liposomal BA for 24, 48, and 72 hours. Cytotoxicity was assessed over a concentration range of 0 – 5.67 mg/mL. Neither BA nor liposomal BA induced significant cytotoxicity at any tested physiological concentration which is modelled to represent the exposure up to six sachets. These findings support the safety of the manufacturer's recommended daily intake of four sachets and suggest that Bet-Aging™ exhibits a favourable *in vitro* safety profile under the experimental conditions evaluated. Nevertheless, further *in vitro* studies are required to validate its safety, efficacy, and anti-ageing mechanisms to support its use as a dietary supplement for healthy ageing.

Keywords: Cytotoxic, HepG2, Bet-Aging, Health Supplement, Safety

WHAT MAKES GOOD DEMENTIA CARE? PERSPECTIVES OF PERSONS WITH DEMENTIA AND FAMILY CAREGIVERS

Sharifah Munirah Syed Elias, Nur Fatihah Shuhada Ismail, Aniawanis Makhtar, Mohd Aznan Md Aris, & Mohd Zulkifli Mohamad Zahir

Abstract

Dementia has emerged as a significant public health concern in Malaysia due to the rapid aging population and increasing demand for long-term care. As formal dementia care services remain limited, the responsibility of care is predominantly undertaken by family caregivers, resulting in substantial challenges for both persons with dementia (PwD) and their family caregivers. Understanding the lived experiences, care needs and preferences of both groups is essential in delivering person-centered dementia care. This study aimed to explore experiences, care needs, and preferences from the perspectives of PwD and family caregivers. A qualitative study design was conducted using purposive sampling across healthcare settings in Kuantan, Pahang. A total of 21 participants, comprising six PwD and 15 family caregivers were recruited in this study. Data were collected through semi-structured interviews. All interviews were audio recorded. and analysed using Naeem's six-step thematic analysis framework. Ethical approval was obtained from the relevant institutional and medical research ethics committees. Four themes emerged: (i) Transitioning into the caregiving role; (ii) Caregiving demands; (iii) The mixed impact of caregiving; (iv) Adaptive care preferences. Dementia caregiving is a complex and progressive process that affects both PwD and family caregivers across physical, psychological, social, and spiritual aspects. The findings highlight the importance of person-centered dementia care that supports the well-being and resilience of both PwD and family caregivers throughout the caregiving journey.

Keywords: Person with Dementia, Family Caregiver, Dementia Care, Qualitative Study

MUSCLE POWER MATTERS: ASSOCIATIONS WITH FRAILTY, PHYSICAL PERFORMANCE, AND BODY COMPOSITION AMONG OLDER PERSONS

Raja Nurzatul Efah Raja Adnan, Hazwan Mat Din, Siti Nor Hadirah Azman, Najwa Najihah Mohd Azmi, Ajun Chin, & Nur Hafidah Ishak

Abstract

Traditionally, muscle strength has been the primary focus in assessing physical performance among older persons. However, recent studies revealed that muscle power deteriorates faster with ageing. Despite its importance in maintaining physical performance, research exploring muscle power among the older population remains limited. Therefore, this study aimed to examine the associations of muscle power with frailty, physical performance, and body composition among older persons. This study was a cross-sectional study conducted among community-dwelling older persons. Frailty was assessed using the Fried Phenotype Frailty. The time to complete Sit-to-stand (STS), Timed up and go (TUG) and 3 metres walk tests were measured in seconds. Skeletal muscle mass (SMM), Percent body fat (PBF), Skeletal muscle index (SMI), Body fat mass (BFM) and Visceral fat (VF) were assessed by using a body composition analyser. Absolute, relative, and allometric muscle power were derived from STS by using formula calculation. The associations between muscle power and the study variables were examined using multiple linear regression analysis. A total of 92 older persons participated in the study, with a mean age of 69.97 (SD = 0.56) years, and 67.4% were female. The mean allometric, relative, and absolute muscle power values were 31.23 (SD = 0.97), 1.14 (SD = 0.03), and 75.07 (SD = 2.37), respectively. In the adjusted regression analyses, higher allometric and relative muscle power were significantly associated with lower frailty scores and better physical performance, as indicated by shorter TUG and 3 metres walk times. Absolute muscle power also showed significant inverse associations with frailty and physical performance. Regarding body composition, absolute muscle power was positively associated with SMM and SMI, whereas relative muscle power was inversely associated with PBF, BFM, and VF after adjustment. Overall, higher muscle power was consistently associated with more favourable frailty, functional performance, and body composition outcomes.

Keywords: Muscle Power, Frailty, Physical Performance, Body Composition, Older Adults

DEVELOPMENT, VALIDATION AND EVALUATION OF A MULTIDISCIPLINARY INTEGRATED CARE PROTOCOL FOR THE IDENTIFICATION AND MANAGEMENT OF MALNUTRITION-SARCOPENIA SYNDROME IN HOSPITALIZED OLDER ADULTS: A STUDY PROTOCOL

Aliza Haslinda Hamirudin, Saminur Fauzan, Siti Salwana Kamsan, Wan Fathin Fariza Wan Mahmood, Nor Syuhaidah Udin, & Mohd Zulkifli Mohamad Zahir

Abstract

The accelerating rate of Malaysia's older adult population leads the nation to face a threat of malnutrition and sarcopenia, particularly among hospitalized older adults. Despite the dual threat of malnutrition-sarcopenia syndrome in this at-risk population, current practices in Malaysian hospitals lack a multidisciplinary approach due to the absence of integrated care protocol to timely identify and manage this condition. This contributes to further downward trajectory of nutritional and health status, which can become detrimental. This research aims: 1) To develop and validate a multidisciplinary integrated care protocol for the identification and management of malnutrition-sarcopenia syndrome (MSS) in hospitalized older adults at a tertiary public hospital, 2) To assess health outcomes based on the newly developed MSS protocol on nutritional status, sarcopenia, functional status, length of hospital stay, quality of life, cognitive function, mortality and readmission among hospitalized older adults, and 3)To evaluate the feasibility and acceptability of the MSS protocol from the perspectives of patients and healthcare providers. This study will be performed in three phases, with each phase linked to a specific research objective. Phase 1 involves protocol development via scoping review, followed by content and face validation. Phase 2 involves a cross-sectional study among hospitalized older adults (n=273) to identify associations and predictors of MSS. Phase 3 will assess feasibility and acceptability of the protocol from the perspectives of both patients and healthcare providers using a mixed-methods approach. Ultimately, this study aims to improve healthcare delivery by establishing a validated, integrated care protocol at a tertiary public hospital with the potential to transform MSS policy within Malaysia's broader healthcare system. This will set a precedent for other hospitals in MSS care, which could enhance patients' health outcomes through coordinated care by dietitians, geriatricians, physiotherapists and other essential healthcare professionals. Funding: Institutional Research Grant 2024 (SRG25-130-0130) and InBody Asia.

Keywords: Malnutrition-Sarcopenia Syndrome, Older Adults, Multidisciplinary Care, Hospital Inpatients, Integrated Care Protocol

PSYCHOLOGICAL AND DIGITAL DETERMINANTS OF MHEALTH ADOPTION AMONG OLDER ADULTS: THE MEDIATING ROLE OF SELF-EFFICACY

Golaleh Karbasi, Rahimah Ibrahim, Siti Anom binti Ahmad, Ghobad Moradi, Puvaneswaran Kunasekaran, Nur Hafidah Ishak, & Mohd Nazim Mohtar

Abstract

Mobile health (mHealth) technologies offer significant potential in improving access to healthcare and supporting health management among older adults. However, mHealth adoption among older adults has been consistently low, influenced by digital capabilities and psychological factors. Digital literacy may facilitate technology use, while technology anxiety may create psychological barriers to adoption intention. The mechanisms through which these factors influence mHealth adoption intention, particularly the role of self-efficacy, are still poorly understood. This study examined the direct and indirect effects of digital literacy and technology anxiety on mHealth adoption intention among Iranian adults aged 50 years and older, with self-efficacy as a mediating variable. A cross-sectional study was conducted among 500 adults aged 50 years or older, selected from health centres in Tehran province using stratified cluster sampling. Data were collected using a structured questionnaire including validated measures of digital literacy (MDPQ-16), technology anxiety, self-efficacy, and mHealth adoption intention. Partial least squares structural equation modelling (PLS-SEM) was performed using Smart-PLS 4.0. Digital literacy positively predicted self-efficacy ($\beta = 0.602$, $p < 0.001$) and adoption intention ($\beta = 0.227$, $p < 0.001$). Technology anxiety negatively affected both self-efficacy ($\beta = -0.177$, $p < 0.001$) and adoption intention ($\beta = -0.099$, $p < 0.05$). Self-efficacy partially mediated the relationships between digital literacy, technology anxiety, and adoption intention. The model explained 50.7% of the variance in adoption intention ($R^2 = 0.507$). The findings suggest that self-efficacy represents an important psychological mechanism linking digital literacy and technology anxiety with mHealth adoption intentions among older adults. Interventions that target strengthening digital skills and reducing technology anxiety are essential to promote mHealth adoption in older populations.

Keywords: Health, Older Adults, Self-Efficacy, Digital Literacy, Technology Anxiety

TOWARDS A NATIONAL AGE-FRIENDLY HOSPITAL FRAMEWORK FOR MALAYSIA

Zahira Zohari, Hakimah Sallehudin, Rahimah Ibrahim, & Tan Maw Pin

Abstract

Malaysia is expected to become an ageing nation by 2030, with 17% of inpatient admissions already being older adults (MOH, 2023). Hospitals are not designed around their needs. By 2026, only 94 consultant geriatricians exist nationwide, half in Ministry of Health hospitals and the rest in university or private hospitals. Age-friendly hospital care in Malaysia largely resulted from local efforts, not a national standard. Malaysia successfully adopted the Baby-Friendly Hospital Initiative, accrediting all government hospitals after a national decision, but this was not extended to older persons. Narrative review of peer-reviewed and grey literature (including MOH and WHO reports), 2010–2025, synthesised across five domains: policy; hospital models; challenges; international comparators; recommendations. Malaysia has accumulated substantial policy intent, but none is a structured ward-level care model. Geriatric services are uneven, concentrated in urban tertiary and university centres, lacking national certification, standardised protocols, outcome registry, or dedicated budget. Workforce shortages, geographic inequalities, and fragmented care recur across all five domains reviewed. These recurring gaps trace to one structural failure: no named framework, no hospital-level owner, no dedicated funding. Four parallel priorities follow. First, a national age-friendly hospital framework with tiered designations based on the 5Ms (mobility, mind, medications, multicomplexity, and matters most), allowing hospitals to enter at their current capacity. Second, baseline geriatric competence across all disciplines, since specialists alone cannot close the gap. Third, integrated care pathways, with structured discharge and formal hospital-community liaison. Fourth, a national geriatric registry and quality indicators, without which none of the above can be sustained. The ageing nation threshold is four years away.

Keywords: Age-Friendly Health Systems, Geriatrics, Health Policy, Hospitals, Public, Malaysia

SELF-RATED HEALTH AS A MEDIATOR OF INPATIENT VISITS AMONG OLDER ADULTS

Nurul Salwana Abu Bakar, Aidalina Mahmud, & Lim Poh Ying

Abstract

Inpatient visits is common among older adults with health problems. However, the role of self-rated health (SRH) in explaining inpatient visits is poorly understood. This study examined whether SRH mediated the relationship between age, financial health coverage, multimorbidity, chronic pain, health symptoms, and inpatient visits. The findings may help identify older adults at risk of inpatient visits, with intersectionality between self-rated health and healthcare. Data from 2,504 older adults were analysed. Factors associated with inpatient visits were identified using multiple logistic regression. Hayes mediation approach was conducted. The direct, indirect, and total effects were examined using generalised structural equation modelling (GSEM), and 5,000 bootstrap samples were used to report bias-corrected confidence intervals. Poor SRH was reported by 47.2% of respondents, and 8.9% reported having had inpatient visits within the past year. Age, financial health coverage, multimorbidity, chronic pain, health symptoms, and poor SRH were significantly associated with inpatient visits. Older adults aged 75–84 years and those aged 85 years and above had higher odds of inpatient visits compared with those aged 60–74 years. In separate mediation analyses, SRH significantly mediated the relationship between chronic pain and inpatient visits, with an indirect effect of (unstandardised ß = 0.55, 95% bias-corrected CI: [0.01, 1.09], $p = 0.04$). A significant mediating effect was found between health symptoms and inpatient visits, with an indirect effect of (unstandardised ß = 0.39, 95% bias-corrected CI: [0.07, 0.88], $p = 0.01$). The findings suggest that SRH may explain part of the relationship between health problems and inpatient visits when these factors are examined separately. Further longitudinal studies are needed to confirm the direction of these relationships and determine whether improving SRH can reduce inpatient visits, as an intersectionality between social and medical areas.

Keywords: Older Adults, Self-Rated Health, Inpatient, Mediation Analysis, Chronic Pain

MILD BEHAVIOURAL IMPAIRMENT AS A PREDICTOR OF DEMENTIA: A SYSTEMATIC REVIEW OF CLINICAL, NEUROBIOLOGICAL, AND PSYCHOSOCIAL EVIDENCE (2016–2026)

Ahmad Shahril Ab. Halim, Norili Farhana Ahmad Saberi, Asma Parveen, Heba Almassri, & Asrenee Ab Razak

Abstract

Dementia remains a leading cause of disability among older adults. Early detection still depends heavily on cognitive markers, which tend to appear only after the disease has already progressed. Mild behavioural impairment (MBI), the later-life onset of persistent neuropsychiatric symptoms in people without dementia, has drawn growing interest as a possible early marker preceding cognitive decline. Most existing reviews examine prevalence alone or focus narrowly on a small set of biomarkers. A synthesis bringing together the clinical, biological, and psychosocial dimensions of MBI as a predictor of dementia, across different settings, tools, and study designs, has been lacking. This review followed the PRISMA 2020 framework. In June 2026, Scopus and Web of Science were searched using the terms "mild behavioural impairment," "predictor," and "dementia," limited to peer-reviewed articles published between 2016 and 2026. Fifty-four studies met the inclusion criteria after screening. Fifty-two were retained following quality appraisal and carried forward for thematic analysis. Three themes emerged. The first characterizes MBI itself: its clinical presentation, its prevalence across different cognitive states, and its rate of conversion to dementia. The second addresses the underlying biology, drawing on fluid biomarkers, structural neuroimaging, and genetic evidence linking MBI to Alzheimer's disease pathology. The third covers risk factors and downstream consequences, including what makes MBI more likely to develop, and its impact on functional decline, quality of life, and caregiver burden. Conversion to dementia reached 71.5% in some cohorts, and MBI was consistently associated with Alzheimer's-specific biomarkers such as plasma p-tau217 and CSF tau/amyloid ratios. The evidence indicates MBI is a meaningful predictor of dementia, both clinically and biologically. Screening for MBI in older adults, particularly in psychiatric and memory clinic settings, may be worth incorporating into routine practice to support earlier intervention.

Keywords: Mild Behavioural Impairment, MBI, MBI-C, Dementia, Alzheimer's Disease, Neuropsychiatric Symptoms, Predictor, Cognitive Decline, Biomarkers, Systematic Review

BRIDGING CLINIC AND COMMUNITY: A CASE STUDY ON HEARING SCREENING INITIATIVES FOR OLDER ADULTS

Mohd Fadzil Nor Rashid, Aw Cheu Lih, Mahamad Almyzan Awang, Maziah Romli, Mohd Normani Zakaria, Wan Najibah Wan Mohamad, & Hasrul Hosshan

Abstract

As the global population ages, the percentage of adults over 60 is rising rapidly across both developed and developing nations. Within this demographic, hearing loss is a highly prevalent chronic condition, affecting between 37% and over 80% of individuals depending on age and geographic location. This case study investigates the rollout of a community-based hearing screening program targeting older adults in remote settings. The project screened a cohort of 60 elderly participants (aged 60 to 79; Mean = 65, SD = ±4.99). Prior to testing, 53% (n = 32) self-reported experiencing hearing difficulties. Screening results revealed that 55% (n = 33) of the participants failed the assessment and were subsequently referred for comprehensive diagnostic audiological evaluations. Our analysis revealed that hearing loss is remarkably common among those 60 and older, with incidence escalating alongside advancing age, chronic health issues, lifestyle choices, and demographic variables. When left unmanaged, hearing loss is linked to cognitive decline, depression, social withdrawal, and a diminished quality of life. Ultimately, to ensure the success of community-oriented healthcare initiatives, systemic barriers—such as financial costs, social stigma, and restricted access to services—must be actively addressed.

Keywords: Hearing Loss, Hearing Screening, Elderly, Pure Tone Audiometry, Community

NEUROPROTECTIVE EFFECTS OF MORINGA OLEIFERA CHEWABLE TABLETS ON COGNITIVE DECLINE IN STZ-INDUCED SPRAGUE-DAWLEY RATS: EVIDENCE FROM Y-MAZE BEHAVIORAL ASSESSMENT

Yus Aniza Yusof, Nurjuliana Mokhtar, Nur 'Aliaa Abd. Rahman, Hafandi Ahmad, Aswir Abd Rashed, Mohammad Adi Mohammad Fadzil, Muhamad Faiz Othman, Wan Hamirul Bahrin Wan Kamal, Saifullizan Mohamad, Anee Suryani Sued, Siti Suhaila Md Izah, Ng Yen, Nur Fariza Mohd Kamil, Siti Aminah Muhamad, Hishar Hassan, & Syaiful Hizam Kamaruddin

Abstract

Cognitive decline associated with ageing and neurodegenerative disorders is closely linked to cholinergic dysfunction, oxidative stress, and impaired neuronal function. Acetylcholinesterase (AChE) inhibition remains an important therapeutic strategy for improving cognitive performance. *Moringa oleifera* is a natural source of polyphenols and flavonoids with reported antioxidant and neuroprotective properties, including the potential to modulate AChE activity. This study aims to evaluate the neuroprotective effects of *Moringa oleifera* chewable tablets in a streptozotocin (STZ)-induced cognitive impairment model using behavioural and biochemical assessments. Adult male Sprague-Dawley rats were allocated into normal control, STZ-induced control, healthy group and treatment groups. Cognitive impairment was induced by intraperitoneal administration of streptozotocin (STZ), followed by oral administration of *Moringa oleifera* chewable tablets according to the experimental protocol. Working memory was assessed using the Y-maze spontaneous alternation test before and after treatment. Serum acetylcholinesterase (AChE) activity was determined to evaluate cholinergic function and to investigate the potential mechanism underlying the cognitive effects of the intervention. Behavioural performance in the Y-maze and serum AChE activity were evaluated to determine the effect of *Moringa oleifera* chewable tablet supplementation on cognitive function. Comparisons among the experimental groups focused on spontaneous alternation performance as an indicator of working memory together with serum AChE activity as a biochemical marker of cholinergic modulation. The combined behavioural and biochemical findings were used to assess the neuroprotective potential of the intervention. This study investigates the potential of *Moringa oleifera* chewable tablets as a functional food intervention for attenuating STZ-induced cognitive impairment. The integration of Y-maze behavioural assessment with serum AChE analysis provides a comprehensive approach for evaluating cognitive function and cholinergic modulation, contributing to the development of evidence-based nutritional strategies for healthy ageing.

Keywords: Moringa Chewable Tablets, Neuroprotective, Behavioral Assessment, Y-Maze, Cognitive Decline

SILENT MARITAL DISTRESS, VISIBLE POSSESSION: SAKA AS A CULTURAL IDIOM OF DISTRESS IN AN ELDERLY MALAY WOMAN

Nabilah Abdul Rahman, & Asrenee Ab Razak

Abstract

Possession Trance Disorder (PTD) is a dissociative disorder characterised by recurrent episodes of altered consciousness in which an individual's identity is temporarily replaced by a perceived external possessing entity. In the Malay community, such experiences are commonly understood as *Saka*, a culturally embedded belief referring to a spiritual inheritance from deceased ancestors believed to possess traditional healing or shamanic powers. This cultural explanatory model shapes the interpretation and expression of psychological distress. We present the case of an elderly Malay woman from Kelantan, Malaysia, with recurrent possession episodes attributed by her family to *Saka*. Comprehensive psychosocial assessment revealed persistent marital conflict characterised by emotional neglect, interpersonal tension, and difficulty expressing disappointment and frustration towards her husband. Raised within a culture emphasising patience, marital harmony, and emotional restraint, she habitually suppressed her emotional needs. Rather than verbalising her distress, her suffering emerged through culturally meaningful possession experiences accepted within her community. This case illustrates possession as a culturally sanctioned idiom of distress rather than solely evidence of severe mental illness. The episodes provided an indirect and socially acceptable means of expressing emotions that were otherwise difficult to articulate because of cultural and relational expectations. Understanding such presentations requires attention to the interaction between cultural beliefs, gender roles, family dynamics, and psychosocial stressors. Without adequate cultural formulation, clinicians may misinterpret these presentations as primary psychotic disorders, resulting in inappropriate treatment, delayed psychological intervention, and increased stigma. This case highlights the importance of culturally informed mental health assessment in Malaysia. Recognising *Saka* as an idiom of distress enables clinicians to identify underlying emotional suffering and relational conflict while respecting patients' cultural beliefs. Integrating cultural understanding with psychosocial assessment facilitates accurate diagnosis, culturally sensitive care, and effective interventions for older adults presenting with possession-related symptoms.

Keywords: Saka, Possession Trance Disorder, Cultural Idiom of Distress, Major Depressive Disorder, Malay, Culturally Sensitive Psychiatry

EFFECTIVENESS OF THE LOW INTENSITY, REPETITIVE, MULTIMODAL WALKING EXERCISE (GOLDEN STEPS) ON PHYSICAL ACTIVITY LEVEL AMONG THE ELDERLY

Nur Syahirah Mohamad Rosli, Noorismiza Zainudin, Siti Nur Baait Sokran, Muslimah Ithnin, Sakinah Rasli, Nur Hafini Harith, & Nurul'ain Shuhaimi

Abstract

Physical inactivity is a major modifiable risk factor associated with functional decline and chronic diseases among older adults. Regular walking is a simple and cost-effective strategy to promote active ageing and improve physical activity levels. The Golden Steps programme, a low-intensity, repetitive, multimodal walking exercise, was developed to encourage sustainable physical activity among older adults. This study aimed to determine the effect of the Golden Steps walking programme on physical activity levels among older adults. A quasi-experimental pre-test and post-test study was conducted among 38 older adults selected using purposive sampling. The minimum sample size was calculated using G*Power (n = 37). Physical activity levels were assessed before and after the eight-week Golden Steps walking programme using the International Physical Activity Questionnaire (IPAQ). Descriptive statistics and paired samples *t*-test were used to analyse participant characteristics and compare pre- and post-intervention IPAQ scores, with significance set at $p < 0.05$. Among the 38 participants, 23 (60.5%) were male and 15 (39.5%) were female, with the majority being Malay (89.5%). Before the intervention, 10 (26.3%) participants had low physical activity levels and 28 (73.7%) had moderate levels. Following the intervention, low physical activity levels decreased to 4 (10.5%), while moderate levels increased to 34 (89.5%). The paired samples *t*-test demonstrated a significant improvement in IPAQ scores after the walking programme (mean difference = 313.66 MET-min/week, SD = 129.51, $t(37) = 14.93$, $p < 0.001$). The Golden Steps walking programme significantly improved physical activity levels among older adults. This low-intensity, multimodal walking exercise may be an effective community-based approach to promote active ageing.

Keywords: Golden Steps, Walking Programme, Physical Activity, Older Adults, Active Ageing, IPAQ

VIRTUAL REALITY-BASED RELIGIOUS TOURISM AND PSYCHOLOGICAL WELLBEING AMONG MUSLIM OLDER PERSONS: A MIXED-METHODS STUDY PROTOCOL

Mohd Rizal Hussain, Mohd Nazim Mohtar, Nur Hafidah Ishak, Raja Nurzatul Efah Raja Adnan, Ajun Chin, Norhalina Senan, Noorhaniza Wahid, Rahayu A. Hamid, Norhanifah Murli, Che Samihah Che Dalim, Suriawati Suparjoh, & Muhammad Fakri Othman

Abstract

Population ageing presents increasing challenges for maintaining psychological wellbeing among older adults. Among Muslim older persons, participation in religious tourism may be limited by mobility difficulties, financial constraints, and health-related barriers. Virtual reality (VR) offers an innovative means of facilitating immersive religious experiences without requiring physical travel. However, evidence regarding the feasibility, acceptability, and potential psychological benefits of VR-based religious tourism among Muslim older persons, particularly in Malaysia, remains limited. This study protocol describes a mixed-methods pilot intervention designed to explore the feasibility and acceptability of VR-based religious tourism and examine preliminary changes in psychological wellbeing, spirituality, and related outcomes among Muslim older persons. A total of 60 community-dwelling Muslim older persons aged 60 years and above in Batu Pahat, Johor, will be recruited using purposive sampling. Participants will be assigned to two groups based on their previous physical religious tourism (Hajj or Umrah): those who have previously experienced physical religious tourism and those who have never performed physical religious tourism. Both groups will undergo the same immersive virtual reality (VR)-based religious tourism intervention, consisting of sessions twice weekly over two weeks using a purpose-built VR application. Quantitative assessments will examine spiritual wellbeing, happiness, life satisfaction, depressive symptoms, goal attainment, and system usability before and after the intervention. Within-group changes and differences in outcomes between the two groups will be explored using appropriate statistical analyses. Qualitative data from focus group discussions will be analysed thematically to understand participants' experiences, perceptions, and acceptance of VR-based religious tourism. This study will provide preliminary insights into the feasibility and acceptability of VR-based religious tourism among Muslim older persons and explore whether previous physical religious tourism experience is associated with differences in their experiences and responses to the intervention. The findings may inform the development of culturally appropriate VR interventions and guide future larger-scale studies on technology-assisted approaches to spirituality and psychological wellbeing in later life.

Keywords: Virtual Reality, Religious Tourism, Muslim Older Persons, Psychological Wellbeing, Spirituality, Study Protocol, Mixed-Methods Pilot Study

SIMAGEING: PREDICTIVE SPATIAL CAD MODELLING AND AGENT-BASED BEHAVIOURAL ANALYTICS FOR NEXT-GEN MHIT UNDERWRITING

Ferozan Ahmad, & Mohd Nazim Mohtar

Abstract

Traditional health and long-term care insurance underwriting models rely mainly on retroactive demographic data, creating a key operational blind spot regarding the impact of physical design on liability. We present a future generation Medical Health Insurance Technology (MHIT) underwriting paradigm that couples computational architecture with clinical neurology for predictive audits of spatial environmental hazards. The target Population, as structured by the PICO framework, consists of high-risk geriatric and cognitively impaired cohorts, including those with advanced dementia or a history of stroke, in complex public and residential built environments. The suggested Intervention is a new simulation engine combining predictive spatial CAD modeling and neuromorphic agent-based behavioral analytics. The engine reads vector data straight from MicroStation design plans, discretizes the layout into a voxel-based matrix with a computationally efficient Chebyshev distance metric, and spawns autonomous digital agents. The engine adds stochastic noise to the agents' pathfinding loops to mimic biological degeneration of the entorhinal cortex, and employs medical odds ratios to determine localized gait instability, so as to properly reproduce clinical disease. This predictive intervention is in Comparison with typical static building code compliance guidelines and retrospective, historical claim-based actuarial models. The main Outcomes are the quantified path trajectory deviations and the localized absolute fall probability. By coupling with a multi-objective evolutionary layout optimization loop (NSGA-II), the results show that the generative system can automatically improve the geometric floor plan layout, reducing the cumulative cohort fall risk by as much as 42% without violating real-world structural engineering constraints. Ultimately, this paradigm offers underwriters with a comprehensive and proactive spatial auditing tool. This turns static architectural CAD geometry into dynamic behavioral risk telemetry, turning next-gen underwriting from a reactive claims-based paradigm to a predictive precision-targeted risk reduction approach.

Keywords: Medical Health Insurance Technology (MHIT), Predictive Spatial CAD Modelling, Neuromorphic Agent-Based Modelling (ABM), Geriatric Fall Risk Telemetry, Long-Term Care Insurance Underwriting.

THANK YOU